FYUGP B.A. (HONS) IN ECONOMICS / B.A. (HONS.) IN ECONOMICS

(RESEARCH)UNDER THE HONORS SCHOOL SYSTEM FOR THE SESSION 2024-25

Outline of Syllabi and Course Structure for FYUGP B.A.(Hons) in Economics /B.A.(Hons.) in Economics (Research) under the Honors School System for the Session 2024-25 in the Department of Economics

NEP Outlines of Syllabi and Course Structure for various Courses in the Department of Economics 2024-25.

B.A. (Hons.) Semester I (under NEP)

Economics Core Course1: Introductory Microeconomics (4 credits)

Economics Core Course 2: Mathematical Methods for Economics-I (4 credits)

Skill Enhancement Course I: Data Collection and Handling Techniques (3 credits) /Any other course offered by Department of Mathematics / Statistics / Computer Science of the Panjab University of same or more credits

Multidisciplinary Course 1: Fundamentals of Sociology (3 credits)/any other course offered by the Panjab University of same or more credits

Ability Enhancement Compulsory Course (AECC) –I (2 credits) (offered by Department of English)

Value Added Course 1: Indian Economic Thought (2 credits)/any other course offered by the Panjab University of same or more credits

Value Added Course 2: Introduction to data Analysis Using Excel-1 (2 credits)/Any other course offered by Departments of Mathematics / Statistics / Computer Science of the Panjab University of same or more credits

Minor 1
Calculus (4 credits)/any other course offered by Department of Mathematics of the Panjab University of same or more credits

B.A. (Hons.) Semester II (under NEP)

Economics Core Course 3: Introductory Macroeconomics –II (4 credits)

Economics Core Course 4: Mathematical Methods for Economics-II (4 credits)

Skill Enhancement Course 2: Data Analysis for Beginners (3 credits)/ Any other course offered by Departments of Mathematics / Statistics / Computer Science of the Panjab University of same or more credits

Multidisciplinary Course 2: Social Stratification (3 credits)/ Any other course offered by the Panjab University of same or more credits

Ability Enhancement Compulsory Course (AECC) –II (2 credits) (offered by Department of English)

Value Added Course 3: Introduction to Environmental Economics (2 credits)/Any other course offered by the Panjab University of same or more credits

Value Added Course 4: Introduction to data Analysis Using Excel-II (2 credits)/ Any other course offered by Department of Mathematics / Statistics / Computer Science of the Panjab University of same or more credits

Minor 2

Algebra (2 credits) + Differential Equations (2 credits)/Any other course offered by the Department of Mathematics of the Panjab University of same or more credits

B.A. (Hons.) Semester III (under NEP)

Economics Core Course 5: Intermediate Microeconomics-I (4 credits)

Economics Core Course 6: Introductory Statistical Methods for Economic Analysis (4 credits)

Economics Core Course 7: History of Economic Thought (4 credits)

Skill Enhancement Course 3: MATLAB (3 credits) / Any other course offered by Department of Mathematics / Statistics / Computer Science of the Panjab University of same or more credits

Multidisciplinary Course 3: Social Stratification (3 credits)/Any other course offered by the Panjab University of same or more credits

Ability Enhancement Compulsory Course (AECC) –III (2 credits) (Punjabi/Hindi) (Offered by Department of Punjabi/Hindi)

Minor 3
MAT-M-3: Partial Differential Equations and Numerical Analysis (4 credits)/Any other course offered by Department of Mathematics of the Panjab University of same or more credits

B.A. (Hons.) Semester -IV (under NEP)

Economics Core Course 8: Intermediate Microeconomics-II (4 credits)

Economics Core Course 9: Intermediate Macroeconomics (4 credits)

Economics Core Course 10: Introductory Econometrics (4 credits)

Economics Core Course 11: Growth Models (4 credits)

Ability Enhancement Compulsory Course (AECC) –III (2 credits) (Punjabi/Hindi) (Offered by Punjabi/Hindi Department)

Minor 4

Statistics: STAT-M-162: Introductory Probability (4 credits)/Any other course offered by Departments of Statistics of the Panjab University of same or more credits

Course Structure and Syllabi for FYUGP B.A. (Hons.) Economics under the Honours School System for 1st, 2nd, 3rd and 4th Semester for 2024-25 Session

Course Structure for FYUGP B.A. (Hons.) Economics under the Honours School System for 1st, 2nd, 3rd and 4th Semester for the Session 2024-25

Semester-I			
Course	**Course Code**	**Paper**	**Credits**
Economics Core Course 1	ECO DSC 1	Introductory Microeconomics	4
Economics Core Course 2	ECO DSC 2	Mathematical Methods for Economics-I	4
Skill Enhancement Course 1	ECO SEC 1	Data Collection and Handling Techniques	3
Multidisciplinary Course 1	MDSC 1	Fundamentals of Sociology	3
Ability Enhancement Compulsory Course 1	AECC 1	English	2
Value Added Course 1	*ECO VAC 1	Indian Economic Thought	2
Value Added Course 2	ECO VAC 2	Introduction to data Analysis Using Excel-I	2
Minor:			
Mathematics Course 1	**MAT-DSC-101	Calculus	4
Total Credits			24

Semester-II			
Economics Core Course 3	ECO DSC 3	Introductory Macroeconomics	4
Economics Core Course 4	ECO DSC 4	Mathematical Methods for Economics-II	4
Skill Enhancement Course 2	ECO SEC 2	Data Analysis for Beginners	3
Multidisciplinary Course 2	PUB MDSC 2	Basics of Indian Administration	3
Ability Enhancement Compulsory Course 2	AECC 2	Course offered by Department of English	2
Value Added Course 3	*ECO VAC 3	Introduction to Environmental Economics	2
Value Added Course 4	ECO VAC 4	Introduction to data Analysis Using Excel-II	2
Minor:			
Mathematics Course 2	MAT-DSC 151	Algebra and Ordinary Differential Equations	4
Total Credits			24
Semester-III			
Economics Core Course 5	ECO DSC 5	Intermediate Microeconomics-I	4
Economics Core Course 6	ECO DSC 6	Introductory Statistical Methods for Economic Analysis	4
Economics Core Course 7	ECO DSC 7	History of Economic Thought	4
Skill Enhancement Course 3	ECO SEC 3	MATLAB	3
Multidisciplinary Course 3	MDSC 3	Social Stratification	3
Ability Enhancement Compulsory Course 3	AECC 3	Course offered by Department of Punjabi/Hindi	2
Minor:			
Mathematics Course 3	MAT-M-3	Partial Differential Equations and Numerical Analysis	4
Total Credits			24

Semester-IV			
Economics Core Course 8	ECO DSC 8	Intermediate Microeconomics- II	4
Economics Core Course 9	ECO DSC 9	Intermediate Macroeconomics	4
Economics Core Course 10	ECO DSC 10	Introductory Econometrics	4
Economics Core Course 11	ECO DSC 11	Growth Models	4
Ability Enhancement Compulsory Course 4	ECO AECC 4	Course offered by Department of Punjabi/Hindi	2
Minor:			
Statistics Course 1	STAT-M-162	Introductory Probability	4
		Internship/CW/FP	2
Total Credits			24

*** ECODSC 1 (4 credits) + ECOVAC 1 (2 credits) would be offered as Minor-1 (6 credits) to Non Economics Honors students in the Semester-I**

*** ECODSC 3 (4 credits) + ECOVAC 3 (2 credits) would be offered as Minor-2(6 credits) to Non Economics Honors students in the semester-II**

Note:

****1. A 6 credit minor (150 marks) would be treated at par with a 4 credit minor (100 marks) and 150 marks to be normalized to 100 for BA. Thus Maths minor students would have more than 4 teaching hours every week.**

***2. Regarding balancing of BSc BA Minor credits the first year A B.Sc. student who opts for a minor of 4 credits from B.A. will cover the remaining 2 Credits from the VAC of the same subject of 2 credits thus earning total 6 credits in that minor and would have two VAC courses in the first and second semester. In the first semester 1 to 2 VAC courses are allowed.**

Syllabi of B.A. (Hons.) Semester- I (under NEP)

ECO DSC 1: INTRODUCTORY MICROECONOMICS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective:-This course is designed to expose the students to the basic principles of microeconomic theory. The emphasis will be on thinking like an economist and the course will illustrate how microeconomic concepts can be applied to analyze real-life situations.

Course Outcomes: -1. To train them to think like an economist. 2.To rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, in general, a static and partial equilibrium framework.

Pedagogy of the Course Outcomes:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:-

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

  The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

  The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

  Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT- I

Exploring the subject matter of Economics. Why study economics? Scope and method of economics; the economic problem: scarcity and choice; the question of what to produce, how to produce and how to distribute output

Cardinal Utility Analysis.

Ordinal Utility Analysis: The consumption decision - budget constraint, consumption and income/price changes, demand for all other goods and price changes; description of preferences (representing preferences with indifference curves); properties of indifference curves; consumers optimum choice; income and substitution effects.

UNIT- II

Theory of production: laws governing short run and long run.

Producer sequilibrium and expansion path

Cost Theory: Short run and long run cost curves.

The Firm and Industry under Perfect Market Structure: Price and output determination; short run and long run.

UNIT- III

Determinants of individual demand/supply; demand/supply schedule and demand/supply curve; market versus individual demand/supply; shifts in the demand/supply curve, demand and supply together; how prices allocate resources. The basic competitive model. Supply and Demand How Markets Work, Markets and Welfare Markets and competition Elasticity of demand: its measurements, applications and determinants. Consumer surplus; producer surplus and the efficiency of the markets.

Concept of dead weight loss; Govt. intervention in the form of Taxes and Subsidies.

UNIT- IV

Single seller monopoly: Price and output determination

Input Markets Labour and land markets - basic concepts (derived demand, productivity of an input, marginal productivity of labour, marginal revenue product); demand for labour; input demand curves;

shifts in input demand curves; competitive labour markets and firms profit maximisation conditions in input market ; and labour markets and public policy. Derivation of labour supply and savings decision - choice between leisure and consumption.

Essential Readings:-

1. N. Gregory Mankiw. (2007) *Economics: Principles and Applications*. (4th ed.). India edition by South Western, a part of Cengage Learning, Cengage Learning India Private Limited.
2. Karl, E. Case., and Ray C. Fair, (2007). *Principles of Economics (*8th ed.). Pearson Education Inc.

Suggested Readings:-

1. Joseph E. Stiglitz, and Carl E. Walsh. (2007) .*Economics* (4th ed.). W.W. Norton & Company, Inc., New York, International Student Edition.
2. Salvotore. D (2006) *Theory and Problems of Microeconomic Theory* (Sachems series) (3rd ed.) Tata McGraw-Hill Publishing Company Ltd.
3. Salvotore.D. (2007). *Micro economic theory and Applications* (4th ed.)Oxford University Press.

ECO DSC 2: MATHEMATICAL METHODS FOR ECONOMICS–I

Topology of the code	**Major**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment20 marks**
Time	**3 hours**

Course Objective:-This is the first part of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in this syllabus. In this course, particular economic models are not the ends, but the

means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication with which the material is to be taught is indicated by the contents of the prescribed textbook.

Course Outcomes: - 1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2. Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3. Students are able to understand and apply the integration of functions and their simple economic applications. 4. Students are able to understand and apply the Difference Equations and their simple economic applications.

Pedagogy of the Course Outcomes:-80% Class room teaching.

20% Internal Assessment.

Instructions for Paper-setter and candidates:

The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

There shall be **9** questions in all.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT- I

Preliminaries: Essentials of Set Theory, Logic, Mathematical Proofs, Mathematical Induction. The real Number System.

Functions of one real variable: Graphs of functions, Linear, quadratic, polynomial, power, exponential, logarithmic.

Properties of Functions: Shifting of Graphs, New functions from Old, Inverse functions, Graphs of Equations, Distance in Plane.

Limits, continuity and series.

UNIT- II

Differentiation: Differentiation concepts, properties with respect to various operations and applications; second and higher order derivatives: properties and applications.

Convex and Concave Functions and Optimization:
Convex and Concave functions: Properties and applications; Local Extreme Points (Maxima and Minima) and applications.

UNIT- III

Integration: Indefinite Integrals, Definite Integrals, Methods of Integration by Parts, Substitution and partial fractions and simple economic applications

UNIT- IV

Difference equations: Introduction, solution of difference equations up to2^{nd} order, simple economic applications.

Essential Readings:-

K. Sydsaeter and P. Hammond, (2016). *Mathematics for Economic Analysis (11th Edition), Pearson: New Delhi {Chapter1: Section 1.4, Chapter 6: Section 6.1 to 6.7, Chapter 9: section 9.1 to 9.6; Chapter 10: section 10.1 to 10.4; Chapter 11: Section 11.1 to 11.4; Chapter 20: section 20.1 to 20.5}*

K. Sydsaeter, P. Hammond, A. Strom and A. Carvajal (2021). Essential Mathematics for Economic Analysis (6 th edition), Pearson. {Chapter 1: Section 1.1 to 1.4; Chapter 4: section 4.1 to 4.10; Chapter 5: Section 5.1 to 5.5; Chapter 6: section 6.1 to 6.11; Chapter 8: Section 8.1 to 8.6; chapter 9: Section 9.1 to 9.6; Chapter 10: section 10.1 to 10.7}

Suggested Readings:-

Weber, Jean E (1982). Mathematical *Analysis Business and Economic Applications* (4^{th}Ed.) Yamane Taro (1972). Mathematics for Economists.

ECO SEC 1: Data Collection and Handling Techniques

Topology of the code	**Skill Enhancement Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective:- To provide skills to undertake data based research so that students get the competency of executing sample surveys and knowledge about secondary data collection.

Course Outcomes : -It will enable the students to qualify for jobs relating to field investigation in any field of social sciences.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be **7** questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of 3marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT-I

Research: Meaning, objectives, Characteristics, Types, Process and Utility, Research method Vs. Methodology. Research Design-Meaning, Types and Features of Good Research Design. Types of Data: Primary vs Secondary, Cross Section or Micro Data, Time Series Data, Pooling of data, Panel data, Census vs sampled data.

UNIT- II

Sampling – Concepts of Statistical Population, Sample, Sampling Frame, Sampling and non-Sampling Errors, Sample Size. Census and Sampling survey. Characteristics of good sampling.
Sampling Techniques: Simple Random Sampling; PPS Sampling; Stratified Random Sampling; Multistage Sampling (Applications only), determination of sample size.

UNIT-III

Methods of collection of Primary data (observation method, interview method, collection of data through questionnaires- Designing of questionnaire, collection of data through schedules- Designing of schedules).Measurement and scaling, validity and reliability.

Collection of Secondary Data
Selection of appropriate method of data collection.

**** Practical Work: Students are required to formulate a research problem and need to prepare a questionnaire for the same along with appropriate measurement and scaling techniques.

Essential Readings:-

1. Kothari, C.R. (2009). *Research Methods and Techniques*, Wiley Eastern.

Suggested Readings:-

1. Bethlehem, J. (2009). *Applied survey methods: A statistical perspective.* Wiley.
2. Cochran, W. (2008). *Sampling techniques, 3rd ed.* Wiley.
3. Cooper, D., Schindler, P., Sharma, J. (2012). *Business research methods*, *12th ed.* McGraw-Hill.
4. Flick, U. (2012). *Introducing research methodology: A beginner's guide to doing aresearch project*. Sage Publications.
5. Groves, R., Fowler, F., Couper, M., Lepkowski, J., Singer, E., Tourangeau, R. (2009). *Survey Methodology*. Wiley.
6. Kumar, R. (2014). *Research methodology: A step by step guide for beginners*, *4th ed.* Sage Publications.

MDSC 1: FUNDAMENTALS OF SOCIOLOGY

Topology of the code	**Multidisciplinary Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment15 marks**
Time	**3 hours**

Course Objective: - The systematic study of human behaviour and human society is a relatively recent development. The central emphasis of the paper would be to develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to development of a sociological outlook. The course would also discuss the works of founding father which led to the development to sociological thinking.

Course Outcomes: -1. Students are able to study human behavior and human society relative to recent development. 2. Students develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development sociological outlook.

Pedagogy of the Course Work:- 80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment will be of 15 marks.

The paper-setter must put a note in the question paper in this regard.

- There shall be **7** questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of 3marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT – I

Developing Sociological Outlook*:* Definition and Scope; Significance and Uses of Sociology; Relationship of Sociology with other Social Sciences – Anthropology, Economics, Political Science, Psychology, Philosophy and History.

UNIT – II

Basic Concepts: **Society** – Meaning and Characteristics, Theories of Origin, Individual and Society. Community, Association and Institution – Meaning, Characteristics and Differences. Groups – Definition, Characteristics and Classification.

UNIT – III

Culture and Socialization: Culture-Definition, Cultural Lag, Acculturation, Assimilation, Pluralism. Culture and Personality. ***Civilization:*** Definition, Traditional and Modern. Socialization – Meaning, Stages, Agencies.

Essential Readings:-

1. Giddens, Anthony .*Sociology, Cambridge (*4th ed.). Polity Press.

Suggested Readings:-

2. Haralambos, Michael.(2015). *Sociology: Themes and Perspectives*. Oxford.
3. Davis, Kingley.(2002). *Human Society.* Surjeet Publications, Delhi.
4. Maclver, R S & Page Charles. *Society: An Introductory Analysis* (Latested.). Macmillan India Ltd.
5. Johnson, Harry (2003). *Sociology: A Systematic Introduction*. Allied Publishers, NewDelhi.
6. Inkeles Alex. (2004).*What is Sociology? An Introduction to the Discipline and Profession*. Prentice-Hall, India.
7. Bradshaw, York W, et.al. (2001). *Sociology for a New*

Century, Piner Forge Press, California.
8. Johnson, Doley Paul.(1986).*Sociological Theory* (Or later ed.). New York, McMillan.
9. Ritzer, George. (2011).*Sociological Theory* .London (Or latest ed.). McGraw-Hill.
10. Coser Lewis. (2004). *Masters of Sociological Thought*, Rawat Publications, NewDelhi.
11. Rao Shankar C.N. (2012).*Sociology*: *principles with an Introduction to Social Thought*.S. Chand and Company Pvt. Ltd.

ECO VAC-1: INDIAN ECONOMIC THOUGHT

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(2 hours a week for 15 weeks= 01semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Course Objective:- The course aims to enable the students to acquire and demonstrate the knowledge on economic thought with its historical perspective.

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Course Outcomes : - 1. Students through this course become familiar with the historical evolution of economic ideas. They are able to learn the foundational contributions of the classical economists to the economic literature.

2. Students are able to develop understanding about the philosophical foundations of economic analysis of capitalist system. They also learn the theories of imperialism which are important to understand the exploitation process of third world countries by the metropolis nations.

3. Students are able to the origin of marginalist, neoclassical and Austrian school of thought and their main features. This is important to understand the differences and similarities among their ideas. 4. Students are able to learn about the Keynesian revolution in economic theory. They also understand the causes of great depression. Moreover, they learn about the Kaleckian version of macro economics.

Pedagogy of the Course Work:- 80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

• The maximum marks for the paper will be 50. The question paper will be of 40

marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x2=30).

UNIT-I

Pre-independence Indian Economic thought

Kautilya's Arthasastra: The economic functions of the state, Taxation and pricing policy, Theland system

Dadabhai Naoroji and the drain theory: What is the drain?, Criticisms and Long-run relevance

Ranade and the economics of development: Poverty and industrialisation, Agrarian policy, Railway investment.

UNIT-II

Indian Economic thought towards independence

Gokhale and the economics of education

Gandhian economics: Methodological issues, Consumption behaviour: the limitation of wants, Swadeshi; Trusteeship and industrial relations; Charity, leisure and the sanctity of work

Essential Readings:-

1. Dasgupta, A.K. (1993). A History of Indian Economic Thought (1st ed.). Routledge.

Suggested Readings:-

2. Kautilya. &Shamasastry, R. (1967). *Kautilya's Arthasastra*. Mysore, India : Mysore Publishing and Printing House.
3. Naoroji, D. (1901). Poverty and Un-British Rule in India. United Kingdom: S.Sonnenschein.
4. Chandra, B., Ranade, M. G. (1990). Ranade's Economic Writings. India: Gian Publishing House.
5. Gokhale, G. K. (2018). Gopal Krishna Gokhale: Select Speeches and Writings. India: National Book Trust, India.
6. Village Swaraj by M.K. Gandhi. (2021). (n.p.): Prabhat Prakashan.
7. Dasgupta, A. K. (1996). Gandhi's Economic Thought. (n.p.): Taylor & Francis.

ECO VAC-2: INTRODUCTION TO DATA ANALYSIS USING EXCEL-I

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(2 hours a week for 15 weeks= 01semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoning involved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x2=30).

Course Outcomes

After successful completion, the students will be able to understand the use of

MS-Excel in data handling and presenting. The filtering techniques, lookup methods, pivot tabling, graphics presentation, matrix analysis etc. shall sharpen a student in the field of quantitative economics and econometrics.

Unit-I

Introduction to Spreadsheets

Introduction to spreadsheets, reading data, manipulating data. Arithmetic operations and basic functions.

Spreadsheet Functions to Organize Data
Introduction to some more useful functions such as the IF, nested IF, VLOOKUP and HLOOKUP functions in Excel.

Unit-II

Introduction to Filtering, Pivot Tables and Charts

Introduction to the Data filtering capabilities of Excel, the construction of Pivot Tables to organize data and introduction to charts in Excel. Constructing various Line, Bar and Pie charts. Using the Pivot chart features of Excel. Understanding and constructing Histograms and Scatterplots.

References:

Lloyd, Chris, J. (2011). Data Driven Business Decisions, John Wiley & Sons.

Levine, D.M., Stephen, D. and Szabat, K.A. (2016). Statistics for Managers Using Microsoft Excel, Pearson Education.

Day, Alastair L. (2013). Mastering Financial Mathematics in Microsoft Excel, Prentice Hall.

Piascik, Chester (2001). Applied Mathematics with Microsoft Excel, Pennsylvania State University.

Jelen, Bill (2013). Excel 2013: Charts and Grap Microsoft Excel. Que Publishing

MAT-DSC-101: Calculus

Credits: 6 (L=5; T=1; P=0)

Total Marks: 150 (Including Internal Assessment= 30) **Time Allowed for Examination:** 3hrs.

Instructions for the Candidates and Paper Setters

Candidates will be asked to attempt five questions out of nine, carrying equal marks. Question No.1 spread over the whole syllabus will be compulsory.

There will be two questions from each unit and the students will have to attempt one from each unit.

Course Outcome: The main goal of this course is to deliver the basics of sequences and series; differential and integral calculus, for real as well as multivariate functions. It is expected that the students develop a taste of writing proofs, particularly for Unit I and Unit III, rather than applying formulas only.

Unit-I

Differential Calculus: Precise definition of limit, continuity, one-sided limit, limits involving infinity, asymp- totes of graphs, tangents and the derivative at a point, the derivative of a function, extreme values of functions, mean value theorem, monotone functions and the first derivative test, test for concavity, tracing of curves.

(Scope: Sections 2.3 – 2.6, 3.1, 3.2, 4.1 – 4.4 of (A)).

Unit-II

Integral Calculus: Riemann sums, definite integrals, area between curves, volumes using cross sections and cylindrical shells, arc length and areas of surfaces of revolution.

(Scope: Sections 5.1, 5.6, 6.1 – 6.4 of (A)).

Unit-III

Infinite sequences and Series: sequences, infinite series, the integral test, comparison test, the root and ratio test, alternating series, absolute and conditional convergence, power series, Taylor and Maclaurin series, Power series, convergence of Taylor series, application of Taylor series.

(Scope: Sections 10.1 - 10.10 of (A).)

Unit-IV

Polar Coordinates: Polar Coordinates, graphing in Polar coordinates, areas and lengths in Polar coordinates.

Multivariable Functions: Limits and continuity for functions of several variables, partial derivatives, the chain rule, directional derivatives, gradient vectors, tangent planes, extreme values and saddle points. (Scope: Sections 11.3-11.5, 14.2 – 14.7 of (A)).

Essential Textbooks

G. B. THOMAS, M. D. WEIR AND J. R. HASS, Inc., Thomas' Calculus, (12th ed.), *Pearson* Education, 2014.

Further Readings

1. J. L. TAYLOR *Foundations of Analysis,* American Mathematical Society, 2012.
2. S. NARAYAN, *Integral Calculus,* S. Chand and Company Ltd, 2001.
3. M.J. STRAUSS, G.L. BRADLEY AND K. J. SMITH, *Calculus,* (3rd ed.), Pearson Education, 2007.
4. H. ANTON, I. BIVENS AND S. DAVIS, *Calculus,* (7th ed.), John Wiley and Sons, 2002.
5. R. COURANT AND F. JOHN, *Introduction to Calculus and Analysis (Volumes I & II),* Springer-Verlag, 1989.

AECC 1: ABILITY ENHANCEMENT COURSE

Course Title: AEC–I: Language Skills: Listening and Speaking
Course Code: ENG (AEC)—030
Typology of Course: AEC
Course Duration: 60 hours of practical in one semester (4hr/week)
Credits: 2 credits
Maximum Marks: 50 marks (30 Theory+15 Practical+5 Internal Assessment)

Learning Objectives of the Course:

- To understand the structure of language and how language varies over time, across social situations and social groups
- To be proficient in articulating thoughts and communicating ideas in English language
- To participate in debates and discussions where ideas are tested and sharpened and honing the skill of public speaking
- Students should be able to hold meaningful conversations
- Student should be able to effectively use English language to meet professional and personal goals

Pedagogy of the Course:

a) The paper requires 4 hours of practicals per week during which students will be taught and explained the theory of communication through practical and hands-on training in various aspects of communication.

b) The practical evaluation, which carries 15 marks, will be done by an external examiner either from the same institution/department or from outside.

Course Content:

Unit I: Sounds of English (**Text**: *English Language Skills: A Practical Approach*, Cambridge).

Unit 2: Language Use in Various Situations (**Text**: *English Language Skills: A Practical Approach*, Cambridge).

Unit 3: Art of Public Speaking (**Text**: *Communication Skills* by Sanjay Kumar & Pushp Lata. OUP 2023)

a. Group Discussion

b. Interview

Suggested Readings:

- *Fluency in English Part II*, Oxford University Press.
- *A Textbook of Phonetics for Indian Students*, T. Balasubramian, Macmillan. *Business English*, Pearson
- *Language, Literature and Creativity*, Orient Black Swan.
- *Language through Literature*, Gauri Mishra, Ranjana Kaul, Brati Biswas (Eds), Primus Books.
- *English Communication,* Macmillan Education.
- *Enrich Your English*, S.R. Inthira and V. Saraswathi, Oxford University Press.
- *Communication and Language Skills*, Sanghita Sen, et al, Cambridge University Press.
- *English and Communication Skills I*, Tasneem Shahnaaz, C. Savitha and Cherlyl Rao, Cambridge University Press.
- *Practical English*, K.K. Singh, APH Publishing Corporation.

Testing
Scheme:

Max. Marks: 50 marks
Theory: 30 marks
Practical: 15 marks Internal Assessment: 05 marks
Time: 1.5 Hours

Note- Question no. 1 is compulsory. Attempt any four questions from Q 2to 7.

Q.1. Short notes. **Two out of five** to be attempted in not more than 150 words each. (10 marks)

Q2. to Q.7. from the prescribed syllabus of Listening and Speaking. **Four out of six** questions to be attempted in not more than 150 words each. (20 marks)

Syllabi of B.A. (Hons.) Semester- II (under NEP)

ECO DSC 3: INTRODUCTORY MACROECONOMICS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregated analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "Introductory Macroeconomics" equips the students at the undergraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Course Outcomes

After completion of the course, a student must be capable of understanding various measures of National income and accounting. Also the paper shall inculcate the fundamental macroeconomic modelling abilities among students to handle macro-economic relationships for their empirical testing. After completion of course, the candidate shall be specifically able to identify the determinants of consumption and investment functions along with the measures of national income; a major cause of variation in two functions.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry 15 marks ($15 \times 4 = 60$).

Unit-I
(Measurement of National Income)

Conceptual Introduction: Aggregates of National and Personal income. Measure of disposable income. The measurement of these aggregates at market and constant prices.

Methods of National Accounting: Expenditure, Value Added, and Income Methods. Mixed methods of national accounting.

Social Accounting: Techniques of Maintaining National Accounts such as Double Entry system, Social Accounting Matrix, National Accounting Equations. The Input Output Tables as base of Social Accounting Matrix. Introduction to Supply and Use Tables.

Unit-II

(Fundamental Theories of Income and Employment Determination)

***Income and Employment Determination*:** Integrated Classical and Keynesian Models of Income and Employment Determination; commodity, money (including bond market of Keynes), and labour markets.

Wage-Price Flexibility and Automatic Full Employment: Classical Versus Keynesian Approach.

Unit-III

(Consumption Theories)

Consumption and Consumption Function: Keynes Consumption and saving functions under Psychological law of consumption, Consumption Puzzle: Absolute Income hypothesis, Relative Income hypothesis, Permanent Income hypothesis and Life Cycle Hypothesis.

Consumption under Uncertainty: Random Walk Hypothesis; Interest Rate and Saving; Consumption and Risky Asset: Consumption CAPM.

UNIT-IV

(Investment Theories)

Investment and Investment Function: Type of Investment, Role of investment using Investment Multiplier, Classical and Keynesian Theories of Investment, Accelerator Theory of Investment, Neo-Classical Theory of Investment and Tobin's-q Theory of Investment. Effects of Uncertainty, Kinked and Fixed Adjustment Costs, Investment in the Housing Market.

Recommended Readings:

Ackley, G. (2007). Macroeconomic Theory***. Macmillan.***

Allen, R.G.D. (1967). Macro-Economic Theory: A Mathematical Treatment***. Palgrave Macmillan.***

Beckerman, W. (1980). Introduction to National Income Analysis, *Littlehampton Book Services Ltd.*

Blanchard, O. (2009). Macroeconomics***. New Delhi: Pearson Education Inc.***

Blanchard, O.J. & Fischer. (1996). Lectures on Macroeconomics***, Delhi: PHI Learning Pvt. Ltd.***

Branson, W. H. (1989). Macro Economic Theory and Policy ***(3rd ed.). Harper Collins.***

Cooper, R., Edey, Harold C., and Peacock, A.T. (2003). National Income and Social Accounting. *Routledge.*

D'Souza, E. (2008). Macroeconomics. ***Pearson Education.***

Dornbusch, R., Fisher, S & Startz, R. (2002). Macroeconomics***, (9th ed.). New Delhi: Tata McGraw Hills.***

Levacic, R. & Rebmann, A. (1982). Macroeconomics: An Introduction to Keynesian Neoclassical Controversies***. Macmillan.***

Miller, R. L. & Pulsinelli, R. (1986). Macroeconomics. ***New York: Harper & Row.***

Ola Olsson (2010). Essentials of Advanced Macroeconomic Theory***, Routledge.***

Romer, D. (1996). Advanced Macroeconomics***. McGraw-Hill.***

Uma Datta Roy Choudhury **(2000).** National Income Accounting, *Macmillan*.

Further Readings:

Abel, A. B., Bernanke, B. S. (2007). Macroeconomics. ***New Delhi: Pearson Education Inc.***

Dornbusch, R. (1980). Open Economy Macroeconomics***. New York: Weltwirtschaftliches Archive.***

Frisch, H. (1983). Theories of Inflation***. Cambridge: Cambridge University Press.***

Froyen, R. T. (2002). Macro Economics. ***New Delhi: Pearson Education Inc.***

Gordon, R. J. (2012). Macroeconomics***. Delhi: PHI Learning Pvt. Ltd.***

Heijdra, B. J. (2017). The Foundations of Modern Macroeconomics***. Oxford University Press.***

Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle ***Oxford*****:** Clarendon Press***.***

Lucas, R. (1981). Studies in Business Cycle Theory***, Massachusetts: MIT Press Cambridge.***

Makin, A. J. (2002). International Macroeconomics***. Princeton Hall.***

Mankiw, N. G. & Romer, D. (1991). New Keynesian Economics. MIT, 2 Volumes.

Mathews, R. C. O. (1964). The Trade Cycle. ***Cambridge: Cambridge University Press.***

Pentecost, E. (2000). Macroeconomics – An Open Economy Approach. ***MacMillan Press Ltd.***

Raghbendra, J. (1964). Contemporary Macroeconomics: Theory and Policy. ***New Age International Publishers.***

Sherffrin, S. (1999). Rational Expectations. ***Cambridge University Press.***

Snowden, B & Vane, H. R. (2005). Modern Macroeconomics: Its Origin, Development and Current Stat., ***UK: Edward Press.***

Wendy, Carlin/Soskice, D. (2007). Macroeconomics: Imperfections, Institutions and Policies***. Oxford University Press.***

ECO DSC 4: MATHEMATICAL METHODS FOR ECONOMICS –II

Topology of the code	**Major**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester),T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20marks**
Time	**3 hours**

Course Objective :-This is the first of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in this syllabus. In this course, particular economic models are not the ends, but the means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication at which the material is to be taught is indicated by the contents of the prescribed textbook.

Course Outcomes : - 1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2. Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3. Students are able to understand and apply the integration o functions and their simple economic applications. 4. Students are able to understand and apply the Difference Equations and their simple economic applications.

Pedagogy of the Course Outcomes: - 80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT- I

Differential Equation*:* Introduction; Solution: Variable Separable Case, Homogeneous Case, Standard Linear Differential Equation, Bernoulli's form, Exact Equation; Solution of Linear differential Equation with Constant Coefficients; Simple Application Questions.

UNIT-II

Linear algebra: Vector spaces: algebraic and geometric properties, scalar products, norms, orthogonality; linear transformations: properties, matrix representations and elementary operations; systems of linear equations: properties of their solution sets; determinants: characterization, properties and economic applications.

UNIT- III

Functions of several real variables: Geometric representations: graphs and level curves; differentiable functions: characterizations, properties with respect to various operations and applications; second order derivatives: properties and applications; the implicit function theorem, and application to comparative statics problems; homogeneous and homothetic functions: characterizations and economic applications.

UNIT- IV

Multi-variable optimization: Convex sets; geometric properties of functions: convex functions, their characterizations, properties and applications; further geometric properties of functions: quasi convex functions, their characterizations, properties and applications; unconstrained optimization: geometric characterizations, characterizations using calculus and applications; constrained optimization with equality constraints: geometric characterizations, lagrange characterization using calculus and applications; properties of value function: envelope theorem and economic applications.

Essential Readings:-

K. Sydsaeter and P. Hammond (2002). *Mathematics for Economic Analysis*,

Pearson Educational Asia Delhi.

K. Sydsaeter and P. Hammond, (2016). *Mathematics for Economic Analysis (11th Edition), Pearson: New Delhi*

{Chapter 5, Chapter 12, Chapter 14: Section 14.1-14.3, Chapter 15: Section 15.1-15.3, Chapter 16, Chapter 17, Chapter 18: Section 18.1-18.7, Chapter 21}

Suggested Readings:-

Weber, Jean E.(1982*). Mathematical Analysis Business and Economic Applications* (4thEd.)
Yamane, Taro (1972). *Mathematics for Economists*.

ECO SEC 2: DATA ANALYSIS FOR BEGINNERS

Topology of the code	**Skill Enhancement Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective:-To provide skills to undertake data based research so that students get the competency of executing sample surveys and knowledge about secondary data collection.

Course Outcomes:- It will enable the students to qualified for jobs relating to field investigation in any field of social sciences.

Pedagogy of the Course Work:- 80% Class room teaching.

20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be **7** questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks. (15x3=45).

UNIT-I

Collection of primary data through questionnaire (online (google form)/ face to face)

Errors in surveys: Misunderstanding of question and answers, problem of non-response.

Processing of Data: Questionnaire checking, Editing, Coding, classification, Tabulation, Graphical representation (Bar charts, Pie Charts, line graphs) Data cleaning, Data Adjusting.

UNIT- II

Analysis of Data and Interpretation: Measures of central tendency (Arithmetic mean median

and mode), Dispersion, simple correlation (Karl Pearson and Spearman Rank Correlation) and simple regression.

UNIT- III

Preparation of a Report: Types of Report, Research Report Format, Principles of Writing Reports, Documentation, Footnotes and References & Bibliography.

Use of computers in research.

Ethics and scientific integrity: standards of conduct, privacy in data, Plagiarism.

****Practical Work: Project Report (collection data through questionnaire on a research problem and preparation, analysis and interpretation of data).

Essential Readings:-

1. Bethlehem, J. (2009). *Applied survey methods: A statistical perspective.* Wiley.
2. Cochran, W. (2008). *Sampling techniques, 3rd ed.* Wiley.

Suggested Readings:-

1. Cooper, D., Schindler, P., Sharma, J. (2012). *Business research methods*, *12th ed.* McGraw-Hill.
2. Flick, U. (2012). *Introducing research methodology: A beginner's guide to doing a research project*. Sage Publications.
3. Groves, R., Fowler, F., Couper, M., Lepkowski, J., Singer, E., Tourangeau, R. (2009). *Survey Methodology*.Wiley.
4. Kothari, C.R. (2009). *Research Methods and Techniques*, Wiley Eastern.
5. Kumar, R. (2014). *Research methodology: A step by step guide for beginners*, *4th ed.* Sage Publications.

PUB-MDSC-2

Multi-Disciplinary Courses (MDSC)

Paper: Basics of Indian Administration

Max. Marks: 75
Theory: 68 Marks
Internal Assessment: 07 Marks
Time: 03 Hours

Course objective

The objective of this paper is to acquaint the student with the basics of Indian Administration. The paper would introduce the student with the basic features of Indian Administration along with the functioning of Union and State Executive and functioning of various constitutional bodies like UPSC, SPSC etc.

Course outcomes

1. To develop understating of the basic features of Indian Administration.
2. To generate awareness about the functioning of Union and State Executive.
3. To delineate the relationship between political and permanent executive.
4. To understand the role of constitutional agencies.

Pedagogy of the course

The students will be assessed on the basis of written examination comprising of 68 marks and continuous internal assessment of 07 marks based on class test, academic activities, Attendance, Class assignments and class room participations.

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES:

- For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.

The Paper-Setter must put a note in question paper in this regard.

- The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 10 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 7. Each question will carry 2 marks. Rest of the paper shall contain 3 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 18 marks.

Course contents

Features of Indian Administration President: Election, Powers and Role

Unit-I

Prime Minister: Appointment, Powers and Role

Unit-II

Governor: Appointment, Powers and Role
Chief Minister: Appointment, Powers and
Role Relationship between Political and
Permanent Executive

Unit-III

Union Public Service Commission: Composition and
Functions State Public Service Commission: Composition
and Functions Comptroller and Auditor General:
Appointment and Functions Election Commission of India:
Composition and Functions

Suggested Readings:

Arora, Ramesh K. and Goyal, Rajni (1997) Indian Public Administration: Institutions and Issues. New Age International Publishers: New Delhi

Avasthi, A and Avasthi, A P (2004) Indian Administration. Laksmi Narain Aggarwal: Agra

Fadia, B L and Fadia, Kuldeep (2017) Indian Administration, (New Edition). Sahitya Bhawan: Agra

Basu, D. D (2013) Introduction to the Constitution of India (21st Edition). Lexus Nexus: New Delhi

Chakraborty, Bidyut (2016) Indian Administration. Sage: New Delhi

Ghuman, B S; Monga, Anil and Johal, Ramanjit Kaur (Eds.) (2012) Corruption and Qualityof Governance: Experiences of Select Commonwealth Countries. Aalekh Publishers: Jaipur

Kangle R P (1972) The KautilyaArthshastra. Motilal Banarsidass: New Delhi

Kapur, Devesh; Mehta, Pratap Bhanu and Vaishnav, Milan (Eds.) (2017) Rethinking Public Institutions in India. Oxford University Press: New Delhi

Maheshwari, S R (2000) Indian Administration. Orient Longman: New Delhi
Palmer, N D (1961) Indian Political System.

Sharma, Ashok(2016) Administrative Institutions in India. RBSA Publishers:

Jaipur Sharma, M (2007) Indian Administration. Anmol: New Delhi

Sharma, Prabhu Datta and Sharma, B M(2009) Indian Administration: Retrospect and Prospect. Rawat Publications: Jaipur

ECO VAC-3: INTRODUCTION TO ENVIRONMENTAL ECONOMICS

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours (3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Course Objective: -This paper aims to make aware students about the importance of environment in economics and vice-versa. It helps them to know the ways of sustaining our natural resources by optimally allocating them for future use.

Course Outcomes: -This paper covers topics related to ecosystems, natural resources, biodiversity, conservation environmental pollution and policies.

Pedagogy of the Course Work:-80% Class room teaching.

20% Internal Assessment.

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Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x2=30).

UNIT-I

Introduction: Environmental economics: Nature, Scope, and Significance. Relation between Environment and Economy (inter linkages); Environment Kuznets Curve. Common property resources and the economic problem; Environmental Externalities in Production and consumption. Managing externalities with taxes and subsidies and Property Rights; Coase Theorem.

UNIT -II

Global Environmental Pollution: Air and water pollution, Acid rain, Ozone depletion and Global warming (causes, consequences, and solutions).

Indian Law pertaining to Environmental protection: Environment Protection Act 1986; Air (Prevention and Control of Pollution) Act; Water (Prevention and control of Pollution) Act; wildlife Protection Act; Forest Conservation Act.

Consumerism and Waste management.

Essential Readings:-

1. Kolstad. C.D. (2011). Intermediate Environmental Economics. Oxford University Press.
2. Goodwin, N., Harris, J., Nelson, J., Rajkarnikar, P., Roach, B., & Torras, M. (2019). Principles of economics in context. Routledge, Taylor & Francis Group, New York & London.
3. Bharucha, E. (2021). Textbook for Environmental Studies (Third edition). Orient Blackswan Pvt Ltd.
4. Eugine, T. (2005) .Environmental Economics. Varinda Publications, Delhi.

Further Readings:-

1. Hussen, Ahmed (2004) .Principles of Environmental Economic (second edition), Routledge, Taylor & Francis Group, London & New York.
2. Goodstein, E.S. (2002).Economics and the Environment (third edition) John Wiley &Sons., Inc.
3. Maddala ,G.S. (2004). Microeconomics; Theory and Practice. McGraw-Hill Education.

ECO VAC-4: INTRODUCTION TO DATA ANALYSIS USING EXCEL-II

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(2 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoning involved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student- teacher interactions.

Instructions for Paper-setter and candidates:

• The maximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.
• There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x2=30).

Unit-I

Macro Recordings and Mathematical analysis
Introduction to Excel Macros. Using Excel-Macro recording for data analyses (without using VB), Solution of matrices.

Unit-II

Statistical Analysis
Central Tendency and Dispersion measures, Regression analysis, Correlation analysis.

References:
Lloyd, Chris, J. (2011). Data Driven Business Decisions, John Wiley & Sons.
Levine, D.M., Stephen, D. and Szabat, K.A. (2016). Statistics for Managers Using Microsoft Excel, Pearson
Education.

Day, Alastair L. (2013). Mastering Financial Mathematics in Microsoft Excel, Prentice Hall. Piascik, Chester (2001). Applied Mathematics with Microsoft Excel, Pennsylvania State University. Jelen, Bill (2013). Excel 2013: Charts and Grap Microsoft Excel. Que Publishing.

Suggestions:

1. The 'spreadsheet operations' should be mentioned specifically. Some books refer this topic as 'arithmetic operation', while others explain it by referring to commands such as 'adding column, deleting row and so on.
2. In semester II, there is imbalance related to distribution of topics between two units. In unit I, there is a single topic "Macros", whereas the second unit consists of 5 topics. The outcome would be better in terms of setting a paper, otherwise there is a chance of higher repetition under question paper.
3. In other respects, the syllabus in fine, and covering almost topic required at the stage of bachelor.

MAT-DSC 151: Algebra and Ordinary Differential Equations

Credits: 6 (L=5; T=1; P=0)

Total Marks: 150 (Including Internal Assessment= 30) **Time Allowed for Examination:** 3hrs.

Instructions for the Candidates and Paper Setters

Candidates will be asked to attempt five questions out of nine, carrying equal marks. Question No.1 spread over the whole syllabus will be compulsory.

There will be two questions from each unit and the students will have to attempt one from each unit.

Objective: The objectives are to introduce a system of linear equations and their solutions, polynomials and their roots and matrix theory. This course will acquaint students to algebra prior to the introduction of more abstract concepts like groups, rings and fields and to exhibit the techniques for obtaining solutions to ordinary differential equations, the basic ideas and the theory behind those techniques. The emphasis will be to combine the abstract concepts with examples in order to intensify the understanding of the subject.

Unit-I

Review of system of linear equations, general theory of system of linear equations, rank of a matrix, row and column ranks of a matrix, system of homogenous and non-homogeneous linear equations, Matrices, Rank of a matrix, Inverse of a Matrix, Elementary linear transformations, Determinants and their properties, Cramer's rule,Cayley-Hamilton Theorem.

Unit-II

A deeper look at Complex Numbers, taking roots of complex numbers, quick review of operations on polynomials, divisors and greatest common divisor, roots of polynomials, fundamental theorem of Algebra(without proof), corollaries of fundamental theorem, Evaluating roots of polynomials of third and fourth degree, bounds of roots, Sturm's theorem, Descarte's rule of signs, approximation of roots.

Unit-III

Origin of Differential Equations, Basic definitions, Family of solutions, Geometric interpretation, Isoclines, Initialand boundary value problem. Basic Existence Theorem (Statement). Equations of Order One, Separation of variables, Exact Equations, Linear Equations, Integrating factors, Bernoulli's equation,

Unit-IV

General linear equations: General solutions, Linear independence of solutions, differential operators, Linear equations with constant coefficients: Auxiliary equation, Non-homogeneous equations: Method of undetermined coefficients, Variation of parameter method, Nonlinear Equations.

(*Scope as in Chapters 2, 3, 4, 5, and 9 of (A), Chapter 1, sections 2.1-2.6, 5.1-5.5, 6.1-6.9, 7.1-7.3, 7.5, 8.3, 8.5, 9.1-9.4 and chapter 16 of (B)*))

Essential Textbook

(A) A. KUROSH *Higher Algebra,* MIR Moscow, 1982.

(B) E. D. RAINVILLE, P.E. BEDIENT AND R. E. BEDIENT, *Elementary Differential Equations,* Eighth Edition, Pearson India, 2016.

Further Readings

1. SHANTI NARAYAN AND P. K. MITTAL, *A textbook of Matrices* S. Chand and Company Limited 2019.

2. S. L. ROSS, *Differential Equations,* 3rd ed., John Wiley and Sons, India 2004.

3. W. E. BOYCE AND R. C. DIPRIMA, *Elementary differential equations and boundary value problems*, Seventh Edition, John Wiley and Sons, Inc., 2001

4. E. A. CODDINGTON, *An introduction to ordinary differential equations,* Prentice- Hall

AECC 2: ABILITY ENHANCEMENT COURSE

Course Title: **AEC– II: Language Skills: Reading and Writing** **Course Code:** ENG (AEC)—031
Typology of Course: AEC
Course Duration: 60 hours of practical in one semester (4hr/week)
Credits: 2 credits
Maximum Marks: 50 marks (30 Theory+15 Practical+5 Internal Assessment)

Learning Objectives of the Course:

- Learn to compose comprehensive and well-structured paragraphs
- Learn to utilize literary terms and vocabulary, critical methods, and various lenses of interpretation in their writing
- To apply rules of grammar
- To learn the formatting and documenting conventions of the discipline

Pedagogy of the Course:

a) The paper requires 4 hours of practical per week during which students will be taught and explained the aspects of reading and writing through practical and hands-on training.

b) For the practical exam, student will be asked to read a text, paraphrase a given text, answer questions and interpret graphs and charts.

c) The practical evaluation, which carries 15 marks, will be done by an external examiner either from the same institution/department or from outside.

Course Contents:

Unit I: Reading (Text: *Interact: A Course in Communicative English*, Cambridge)

1. Close Reading
2. Comprehension
3. Summary
4. Paraphrasing
5. Interpreting Graphs and Charts

Unit II. Writing (Text: *Writing Skills*, Cambridge, 03-57)

1. Paragraph Writing
   (Available in Print and e-book: *Effective Academic Writing I&II,* Alice Savage & Patricia Mayer, OUP (USA, 2006)
2. Note-Making
3. Letter/E-mail Writing
4. Resume Writing
5. Report Writing

Suggested Readings:

- *On Track: English Skills for Success*, Orient Black Swan.
- *Embark: English for Undergraduates*, Steve Hart, Arvind R. Nair and Veena Bharti, Cambridge University Press.
- *Connect: Course in Communicative English*, Debashis Bandyopadhyay and Malathi Krishnan, Cambridge University Press.
- *Enrich Your English*, S.R. Inthira and V. Saraswathi, Oxford University Press.
- *Communication and Language Skills*, Sanghita Sen, et al, Cambridge University Press.
- *Practical English*, K.K. Singh, APH Publishing Corporation.
- *Language, Literature and Creativity*, Orient BlackSwan.
- *The Elements of Style,* William Strunk Jr. and E.B. White, Pearson.
- *On The Art of Writing*, Arthur Quiller-Couch, Notion Press.
- *Fluency in English Part II*, Oxford University Press.

Testing Scheme

Max. Marks: 50 marks Theory: 30 marks
Practical: 15 marks Internal Assessment: 05 marks
Time: 3 hours

Section-A

Q.1. Short Notes. **Two out of five** to be attempted in not more than 150 words each. (10 marks)

Section-B

Attempt two questions out of given four: (10 marks)

Q2. Reading Comprehension
Q.3. Summarise a given passage.
Q.4. Write a paraphrase of the given passage.
Q.5. Interpreting Graphs and Charts

Section-C

Attempt any two questions out of five: (10 marks)

Q.6. Paragraph Writing
Q.7. Note-Making
Q.8. Letter/E-mail Writing
Q.9. Resume Writing
Q.10. Report Writing

Syllabi of B.A. (Hons.) Semester- III (under NEP)

ECO DSC 5: INTERMEDIATE MICROECONOMICS – I

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Description

The course is designed to provide a sound training in microeconomic theory to formally analyze the behaviour of individual agents. Since students are already familiar with the quantitative techniques in the previous semesters, mathematical tools are used to facilitate understanding of the basic concepts. This course looks at the behaviour of the consumer and the producer and also covers the behaviour of a competitive firm.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT- I

Budget constraint; properties of budget set, how budget line changes.

Preference; Indifference curves, Marginal rate of substitution.

Utility; perfect substitutes, perfect complements,

Choice; optimal choice; neutrals and bads, discrete goods, concave preferences.

Demand; income offer curves and Engel curves, price offer curve and demand curve. Revealed preference

UNIT- II

Slutsky equation; the substitution effect, the total change in demand, rates of change.

Buying and Selling; net gross demand, offer curves and demand curves Intertemporal choice.

UNIT- III

Uncertainty and risk aversion
Technology: Production Functions; Homogenous and homothetic;
Cobb Douglas and CES: Properties; elasticity of substitution; Eulers theorem.

UNIT-IV

Perfect competition: Assumptions: theory of a firm under perfect competition demand and revenue; equilibrium of the firm in the short run and long run; long run industry supply curve: Increasing, decreasing and constant cost industries.
Price controls: Price Ceiling; price floors; production import quotas
Welfare: allocative efficiency under perfect competition.

Readings:

1. Scope as in Varian, H.R (2010). *Intermediate Microeconomics, a Modern Approach* (8thed.). W.W. Norton and Company/Affiliated East- West Press (India) given as follows;
   Unit 1 chapters 2 to 7
   Unit 2 chapters 8,9,10
   Unit 3 chapters 12 to 18
   Unit 4 chapters 19, 22, 23

Advanced Readings:

2. Varian.H.R (2010). Intermediate Microeconomics, a Modern Approach (8th ed.)W.W. Norton and Company/Affiliated East-West Press (India).The workbook by Varian and Bergstrom could be used for problems.
3. C. Snyder and W. Nicholson.(2010).Fundamentals of Microeconomics, Cengage Learning(India).
4. Maddala, G.S. and Miller, R. L. (2006). Intermediate microeconomics: theory, issues,applications. McGraw-Hill Companies.
5. Koutsoyiannis, A. (2003). Modern microeconomics.(2nd ed.). Paperback
6. B. Douglas Bernheim and Michael D. Whinston. (2009). Microeconomics, Tata McGraw-Hill (India).
7. Bergstrom, T. C., & Varian, H. R. (1990). Workouts in intermediate microeconomics. WWNorton.
8. Pindyck, Robert S., and Daniel L. Rubinfeld. Microeconomics. Upper Saddle River, N.J.:Pearson/Prentice Hall, 2009.
9. Mankiw, G. (2000). Microeconomics. In *Principles of Microeconomics* (Vol. 1, pp. 3–6,270–272).

ECO DSC 6: Introductory Statistical Methods for Economic Analysis

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective: The Objective of this paper is to make students familiar with the use of statistical tools in analyzing economic problems. The course content contains simple tools and techniques, considered necessary for data collection, presentation and analysis with basic idea of probability concepts which are essential for understanding advance topics related to statistical inference.

Course Outcome: On completion of this course students will be able to: (i) Get familiar with various statistical methods used in summarization and analysis of data. (ii) Understand and able to calculate the measures of central tendency and variability. (iii) Understand about the types of correlations and linear regression for bivariate data and their uses in real life problems. (iv) Understand the basic cocepts of probability to be used in statistical inference.

Pedagogy of the Course Work: 80% Class room teaching.
20% Internal Assessment

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10x2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT – I

Statistics, Data and Statistical Thinking

Science of Statistics, Types of statistical applications, Fundamental elements of Statistics, Types of Data, Collecting data. Classification and Tabulation of data, Graphic and Diagrammatic representation of data. The role of statistics in critical thinking and ethics.

UNIT – II

Numerical Descriptive Measures

Measures of Central Tendency: Mean, Median, Mode, Geometric Mean and Harmonic Mean; *Measures of Variability:* Range, Quartile Deviation, Mean Deviation and Standard Deviation: Interpreting the Standard Deviation.

UNIT – III

Techniques to quantify linear relationship

Correlation and Analysis: Simple Correlation (Karl Pearson's and Rank Correlation); Partial and Multiple Correlation Analyses;
Regression Analysis: Estimation of aRegression line in a Bivariate Distribution, Least Squares Method: Interpretation of Correlation Coefficients with help of Regression Coefficients, Coefficient of Determination

UNIT – IV

Basics of Probability

Sample spaces and events, Definitions of probability (Classical, Empirical and Axiomatic), Unions and intersections, complementary events.

The additive rule and mutually exclusive events, conditional probability, the multiplicative rule and independent events, random sampling and counting rules.

Essential Readings:

Devore, J. (2012). Probability and statistics for engineers, 8th ed. Cengage Learning. (Chapters 1, 2 and 12- 12.1, 12.2 and 12.5-508 to 511 covering unit III)

James McClave, Terry Sincich. (2013). Statistics for Business and Economics, Pearson Publication. (Chapter 1-1.1 to 1.6, Chapter 2-2.1 to 2.6, 2.9 and 2.10 and chapter 3-3.1 to 3.8 covering units I, II and IV)

Suggested Readings

Larsen, R., Marx, M. (2018), An Introduction to Mathematical Statistics and its Applications, Pearson. Gupta, S C .(8th edition). Fundamentals of Statistics, Himalaya PublishingHouse
.Miller, Irwin and Miller, Marylees (2006): John E. Freund's Mathematical Statistics with Applications, (7th Edn.), Pearson Education, Asia.

ECO DSC 7: HISTORY OF ECONOMIC THOUGHT

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Objective:

Evolution of economic ideas is both a response to contemporary economic problems and a self-conscious attempt to refine earlier ideas to integrate them as a part of current social thought.

The objective of this course is to familiarize the students with the historical evolution of economic ideas into the contemporary economics.

The time-tested method of imparting verbal instructions thought lectures would be used. Examples, so far as possible, would be selected from everyday life-experience. The students would be evaluated at the end of semester through subjective type questions/answers (both short and essay type).

Pedagogy of the Course Work: 80% Class room teaching.

20% Internal Assessment Comprising Unit Test and Assignment/term paper.

Instructions for Paper-setter and candidates:

• The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

• There shall be 9 questions in all.

The first question compulsory comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of two marks (10x2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT – I

Scope and Significance of the Study of History of Economic Thought.

Mercantilism: Factors Responsible for the Rise of Mercantilism; The Mercantilist Conception of foreign Trade, State and Money.

Physiocracy: Rise of Physiocracy; Concept of Net Product; Natural Order; Circulation of Wealth.

UNIT – II

Classical School: Adam Smith: Theory of Value; Distribution; Growth; Trade. Ricardo: Theory of Value; Distribution; Trade; Views on Machinery.
Malthus: Market Glut and Glut Controversy. J S Mill: Stationary State.

UNIT – III

Karl Marx: Rise and Fall of Capitalism.
Neo Classical Economics – Main features and contribution. Keynes: Principle of Effective Demand.
Friedman: Introduction of monetary school.

UNIT IV

Evolution of Economic thought post world war –II. 1960's: The decade of Development;

1970's: Inequality debate (Kuznets etal.) Phases of Neo-Liberalism: 1980's and 1990's;

Rise of Neo-Nationalism in 2000's.

Essential Readings:

Jevons, W. S. (1888): The Theory of Political Economy, Macmillan and Co., New York and London. Chapter 1 and 6.

Marshall, (1890): Principles of Economics 8th Edition, Macmillan and Co., New York and London Book III, Chapter 6 and Appendix C51.

Pigou, A.C. (1920): The Economic of Welfare, Macmillan and Co., New York and London, Chapter 20.

Keynes J.M. (1936): The General Theory of Employment, Interest and Money, Reprinted by Atlantic Publication, New Delhi. Chapter 2, 3, 8,9,10 and12.

Kalecki M. (1971): Selected Essays on the Dynamics of Capitalism, University Press, Cambridge.

Friedman, M. (1962): Capitalism and Freedom, The University of Chicago Press, Chicago and London.

Foster, J. B. and: The Endless Crisis, Monthly Review Press. McChesney (2012)

Harvey, D. (2005): A Brief History of Neoliberalism, Oxford University Press, New York.

Steger M.B. and: Neoliberalism: A very Short Introduction, Oxford Roy, R.K. (2010) University Press, New York.

Further Readings:

Screpanti (2006): Outline of History of Economic Thought. Oxford University Press, Calcutta.

Blaug, M (1983): Economic Theory in Retrospect. Vikas Publishing House, New Delhi.

Dasgupta, A K (1986): Epochs of Economic Theory. Oxford University Press, New Delhi.

Collander, D C (1984): Neo-Classical Political Economy, Ballinger, Cambridge, M A.

Mandel, E (1971): Marxist Economic Theory. Upra & Co.

Schumpter, J A (1954): History of Economic Analysis, Oxford University Press, Oxford.

MAT-M-3: Partial Differential Equations and Numerical Analysis

Credits: 6 (L=5; T=1; P=0)

Total Marks: 150 (Including Internal Assessment= 30) **Time Allowed for Examination:** 3hrs.

Instructions for the Candidates and Paper Setters

Candidates will be asked to attempt five questions out of nine, carrying equal marks. Question No.1 spread over the whole syllabus will be compulsory.

There will be two questions from each unit and the students will have to attempt one from each unit.

Objective: To study Ordinary differential equations in more than two variables. To form and solve partial differential equations of first and second orders. Further, to acquaint the students with Numerical approximations, convergence problems, Solution of polynomial and simultaneous equations, Different rules of Numerical integration.

Unit-I

Partial differential equations of the first order:

Origins of first order partial differential equations, Cauchy's problem for first order equations, linear equation of first order, Integral surface passing through a given curve, Surfaces

orthogonal to a given system of surfaces, Non- linear partial differential equation of the first order, Cauchy method of characteristics, Compatible system of first order equations. Charpit's method, Special types of first order equations, Solutions satisfying given conditions, Jacobi's method, Applications of first order partial differential equations.

(Scope: Chapter 2 of (A)).

Unit-II

Partial differential equations of second order: The origin of second order equations, General solution of Linear partial differential equations with constant coefficients. Partial differential equations with variable coefficients, Classification of second order partial differential equations. (Scope: Chapter 2, 3 of (A)).

Method of Separation of variables: Heat conduction problem, Vibrating String problem, boundary value problems, Solution of Laplace equation, Heat equation and Wave equation with method of separation of variables (Cartesian co-ordinates). (Scope: Chapter 14, Section 14.2 of (B))

Unit-III

Algorithms, Convergence, Error Analysis: Relative error, Truncation error, Round off error, order of approxima- tion, order of convergence, Propagation.

Solution of Non-linear equations: Bisection method, Secant Method, Method of false position, Newton Raphson Method, fixed point iteration method, Convergence analysis and order of convergence for all these methods.

Solution of system of linear equations: Gaussian Elimination method, Gauss Jordan, Gauss Jacobi method, Gauss- Seidel Method, LU decomposition Method, Successive-over-relaxation (SOR) iteration methods and their convergence, ill and well conditioned systems, Rayleigh's power method for eigen-values and eigen-vectors.

Unit-IV

Interpolation: Errors in polynomial interpolation, Finite difference operators, Newton's Gregory forward and backward interpolation Formula, and Central difference interpolation

formula: Gauss', Striling's, Bessel's, Everett's. Lagranges interpolation formula and Newton divided difference interpolation formula.

Numerical Integration: Midpoint rule, Trapezoidal rule, Simpson's 1/3 rule, Simpsons 3/8 rule, Composite Trapezoidal rule, Composite Simpson's rule. Boole's Rule and Weddles rule, Romberg integration, Newton Cotes integration formula, Gaussian Quadrature and generalized quadrature. Numerical double integration.

(For Units III and IV, scope as in relevant sections of Chapters 1-4 of (C)).

Essential Textbooks

(A) I.N. SNEDDON, *Elements of Partial differential equations, Dover Publications, Inc. New York, 2006.*

(B) S.L. ROSS, *Differential equations (3rd ed.), John Wiley and Sons, India, 2004.*

(C) M. K. JAIN, R. K. S. IYENGAR AND R. K. JAIN, *Numerical Methods: Problems and Solutions,* New Age International, 2007.

Further Readings

1. MARTHA L ABELL AND JAMES P BRASELTON, *Differential equations with MATHEMATICA (3rd ed.),* Elsevier Academic Press, 2004.
2. W. E. BOYCE AND R. C. DIPRIMA, *Elementary Differential Equations and boundary value problems,* John Wiley and Sons Inc., 2008.
3. TYN MYINT-U AND LOKENATH DEBNATH, *Linear Partial Differential Equations for Scientists and Engineers (4th ed.),* Prentice Hall, 2003.
4. R. L. BURDEN AND J. D. FAIRES, *Numerical Analysis, (9th ed),* Brooks/Cole, Cengage Learning, 2011.
5. C. F. GERALD AND P. O. WHEATLEY, *Applied Numerical Analysis,* Pearson, 2010.
6. R. S. GUPTA, *Elements of Numerical Analysis,* Cambridge University Press, 2015.
7. F. B. HILDEBRAND, *Introduction to Numerical Analysis,* Courier Corporation, 1987.
8. BRIAN BRADIE, *A Friendly Introduction to Numerical Analysis,* Pearson, 2007.
9. URI M. ASCHER AND CHEN GREIF, *A First Course in Numerical Methods, 7th Ed.*, PHI, 2013.
10. JOHN H. MATHEWS AND KURTIS D. FINK, *Numerical Methods using Matlab*, PHI, 2012.
11. S. S. SHASTRY, *Introductory methods of numerical analysis,* PHI, 2012.

MDSC 3: SOCIAL STRATIFICATION

Topology of the code	**Multidisciplinary course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective: -The systematic study of human behavior and human society is a relatively recent development. The central emphasis of the paper would be to develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development sociological outwork. The course would also discuss the works of founding father which led to the development of sociological thinking.

Course Outcomes : - 1. Students are able to study human behavior and human society relative to recent development. 2. Students develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development of sociological terms and themes,

Thus leading to a development of a sociological outwork.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be**7**questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT – I

Social Stratification: Concept - Social Stratification, Inequalities and Differentiation. Elements: Position, Role and Status.

UNIT II

Forms of Stratification and Social Mobility: Forms of Stratification – Caste, Class, Race and Gender. Social Mobility: Definition; Indicators – Social, Economic, Political; Types –

Horizontal and Vertical (Upward and Downward) and Intergenerational.

UNIT – III

Social Change: Social Change – Definition. Factors: Demographic, Technological, Educational, Legislative. Processes of Change: Sanskritization, Westernization, Modernization and Globalization.

Essential Readings:-

1. The New Middle Class in India – A Sociological Analysis, Jaipur, Rawat.
2. Singh, Yogendra (1986). *Modernization of Indian Tradition*, Jaipur, Rawat (or latested.)
3. Srinivas, M N. (2010) .*Social Change in Modern India*, Orient Black Swan Pvt.Ltd.
4. Sharma, K L. (2006). *Social Stratification and Mobility*, Rawat Publication, New Delhi.
5. Abraham Francis, M. (2004). *Modern Sociological Theory – An Introduction*, Oxford University Press, Delhi.

Suggested Readings:-

1. D'Souza, V. S. (1981). *Inequality and Its Perpetuation*, Manohar Publications, New Delhi.
2. Singh, K S. (ed.) (1992). *Ethnicity Caste and People: India and Soviet Union.* Manohar Publications, New Delhi.
3. Singer Milton, Coha Beemard (2001). *Structure and Change in Indian Society*, Rawat Publications, New Delhi.
4. Atal Yogesh.(2006). *Changing Indian Society*, Rawat Publications, New Delhi.
5. Beteille, Andre (2009). Sociology – Essays on Approach & Method. Oxford University Press.
6. *Rao Shankar* C.N. (2012).*Sociology*: principles with an Introduction to Social Thought.S.Chand and Company Pvt. Ltd.
7. *Rao Shankar* C.N. (2012). SOCIOLOGY OF INDIAN SOCIETY.S.Chand and Company Pvt. Ltd.
8. Haralambos, Michael.(2015). Sociology: Themes and Perspectives. Oxford.
9. Ministry of Women and Child Development, GOI. Report of the High Level Committee on Status of Women in India (2015).http://www.wcd.nic.in/documents/hlc-status-women
10. Johnson, Doley Paul.(1986).Sociological Theory (Or later ed.). New York, McMillan.
11. Ritzer, George. (2011).Sociological Theory .London (Or latest ed.). McGraw-Hill.
1. Coser Lewis. (2004). Masters of Sociological Thought, Rawat Publications, New Delhi.
2. Rao Shankar C.N.(2012).Sociology: principles with an Introduction to Social Thought. S. Chand and Company Pvt. Ltd

Compulsory Punjabi
ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ
PUN-128
FYUGPG B.A (Hons.) in Economics/B.A (Hons) in Economics (Research)

Semester- III **ਕ੍ਰੈਡਿਟ-2**

Credit Distribution, Eligibility and Pre-Requisites of the Course

Course title & Code	Credits	Credit distribution of the Course Lecture tutorial practical			Eligibility criteria	Pre-requisite of the course (if any)
PUN-128	2	3	1	0		

Credits: 2 **L/T/P=3/1/0**

Learning objective

- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਨਾਟਕਸੰਬੰਧੀਮੁੱਢਲਾਗਿਆਨਪ੍ਰਾਪਤਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਨਾਟਕਕਾਰਈਸ਼ਵਰਚੰਦਰਨੰਦਾਦੇਨਾਟਕਰਾਹੀਂਨਾਟ-ਕਲਾਤੇਰੰਗਮੰਚਬਾਰੇਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਭਾਸ਼ਾਅਤੇਉਪ-ਭਾਸ਼ਾਦੇਅੰਤਰਸੰਬੰਧਾਂਪ੍ਰਤੀਮੁੱਢਲੀਜਾਣਕਾਰੀਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਦੀਵਿਲੱਖਣਪਛਾਣਤੇਸ਼ਬਦਾਵਲੀਸੰਬੰਧੀਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।

Learning outcomes

- ਇਸਨਾਲਵਿਦਿਆਰਥੀਆਂਅੰਦਰਪੰਜਾਬੀਨਾਟਕਦੀਵਿਧਾਸੰਬੰਧੀਮੁੱਢਲੀਸੂਝਪੈਦਾਹੋਵੇਗੀ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਨਾਟਕਦੇਹਵਾਲੇਨਾਲਪੰਜਾਬੀਨਾਟ-ਕਲਾਤੇਰੰਗਮੰਚਦੀਬਣਤਰਸੰਬੰਧੀਵਿਵੇਕਹਾਸਿਲਕਰਪਾਉਣਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਆਂਵਿਚਭਾਸ਼ਾਤੇਉਪ-ਭਾਸ਼ਾਦੇਨਿਖੇੜਸੰਬੰਧੀਸੂਝਪੈਦਾਹੋਵੇਗੀ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਆਂਵਿਚਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਦੀਵਿਲੱਖਣਪਛਾਣਸੰਬੰਧੀਸਮਝਪੈਦਾਹੋਵੇਗੀ।

Compulsory Punjabi
ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ
PUN-128
FYUGPG B.A (Hons.) in Economics/B.A (Hons) in Economics (Research)

Semester :III

ਕ੍ਰੈਡਿਟ-2
ਨਿਰਧਾਰਤ ਕੋਰਸ ਸਮਾਂ:60 ਘੰਟੇ
ਕੁਲ ਅੰਕ:50
ਲਿਖਤੀ ਅੰਕ:40
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ:10

(ਨੋਟ: ਇਸਕੋਰਸਵਿਚ 2 Unitਹੋਣਗੇ। Unit-1 ਵਿਚਨਾਟਕਨਾਲਸੰਬੰਧਿਤਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ। Unit-2 ਵਿਚਗੁਰਮੁਖੀਲਿਪੀਦੇਮਹੱਤਵਤੇਪੰਜਾਬੀਭਾਸ਼ਾਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਨਾਲਸੰਬੰਧਿਤਸਵਾਲਪੁੱਛੇਜਾਣਗੇਅਤੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ।)

Unit-I

ਈਸ਼ਵਰਚੰਦਰਨੰਦਾਦਾਨਾਟਕ 'ਵਰ-ਘਰ', ਲਾਹੌਰਬੁੱਕਸ਼ਾਪ, ਲੁਧਿਆਣਾ, 1930

(ੳ) ਨਾਟਕਦਾਵਿਸ਼ਾ-ਵਸਤੂਤੇਕਲਾਪੱਖ (ਦੋ ਵਿਚੋਂਇੱਕ)
(ਅ) ਕਿਸੇਪਾਤਰਦਾਪਾਤਰ-ਚਿਤਰਣ (ਦੋ ਵਿਚੋਂਇਕ)
(ੲ) ਨਾਟਕਵਿਚੋਂਲਘੂਪ੍ਰਸ਼ਨ (ਦਸ ਵਿਚੋਂਪੰਜ)

(5+5+10=20)

Unit-II

ਗੁਰਮੁਖੀਲਿਪੀਅਤੇਪੰਜਾਬੀਭਾਸ਼ਾਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂ

(ੳ) ਗੁਰਮੁਖੀਲਿੱਪੀਦਾਮਹੱਤਵਅਤੇਪੰਜਾਬੀਭਾਸ਼ਾਲਈਇਸਦੀਸਾਰਥਕਤਾ (ਦੋ ਵਿਚੋਂਇਕ)
(ਅ) ਭਾਸ਼ਾਅਤੇਉਪ-ਭਾਸ਼ਾ: ਅੰਤਰਸੰਬੰਧਤੇਨਿਖੇੜਾ (ਦੋ ਵਿਚੋਂਇੱਕ)
(ੲ) ਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂ (ਸੱਤਾਵਿਚੋਂਪੰਜ)

(5+5+10=20)

ਨੋਟ: ਲਾਜ਼ਮੀਪੰਜਾਬੀਵਿਸ਼ੇਦੇਕ੍ਰੈਡਿਟ 2 ਹੋਣਗੇਅਤੇਪੇਪਰ 50 ਅੰਕਾਂਦਾਹੋਵੇਗਾ। ਵਿਦਿਆਰਥੀਹਫ਼ਤੇਵਿਚਘੱਟੋ-ਘੱਟਪੰਜਦਿਨਪੜਨਗੇ।

ਪੇਪਰਸੈਟਰ/ਅਧਿਆਪਕ ਲਈ ਹਦਾਇਤਾਂ ਅਤੇ ਪ੍ਰਸ਼ਨ-ਪੱਤਰ ਦੀ ਰੂਪ-ਰੇਖਾ:-

1. ਪ੍ਰਸ਼ਨ-ਪੱਤਰਦੇਕੁੱਲ 50 ਅੰਕਹਨ ਜੋ ਕਿ 2 ਕ੍ਰੈਡਿਟਦੇਬਰਾਬਰਹੈਅਤੇਕੋਰਸਦਾਨਿਰਧਾਰਤਸਮਾਂ 60 ਘੰਟੇਹੋਵੇਗਾ।
2. ਲਿਖਤੀਪ੍ਰਸ਼ਨ-ਪੱਤਰ 40 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।

3. ਇੰਟਰਨਲਅਸੈਸਮੈਂਟਦੇ 10 ਅੰਕਹੋਣਗੇ।

4. ਪੇਪਰਦਾਸਮਾਂਤਿੰਨਘੰਟੇਹੋਵੇਗਾ।

ਯੂਨਿਟਅਤੇਅੰਕਾਂਦੀਵੰਡ

1. ਯੂਨਿਟਇਕਵਿਚ 'ਵਰ-ਘਰ' ਨਾਟਕਵਿਚੋਂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਹਯੂਨਿਟ 20 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।

(ੳ) ਇਸਭਾਗਵਿਚਨਾਟਕਦੇਵਿਸ਼ਾ-ਵਸਤੂਅਤੇਕਲਾਪੱਖਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਅ) ਇਸਭਾਗਵਿਚਪਾਤਰ-ਚਿਤਰਣਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ੲ) ਇਸਭਾਗਵਿਚਨਾਟਕਸੰਬੰਧੀਲਘੂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਸਯੂਨਿਟਵਿਚਕੁਲ 14 ਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਪਹਿਲੇਚਾਰਸਵਾਲਾਂਵਿਚੋਂ ਦੋ ਕਰਨੇਹੋਣਗੇ, ਜੋ ਪੰਜ-ਪੰਜਅੰਕਾਂਦੇਹੋਣਗੇ। ਬਾਕੀ ਦਸ ਲਘੂਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂਪੰਜਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਕਿ ਦੋ-ਦੋ ਅੰਕਦੇਹੋਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁੱਲਅੰਕ 20 ਹੋਣਗੇ।

2. ਯੂਨਿਟ ਦੋ ਵਿਚਗੁਰਮੁਖੀਲਿਪੀਦੇਮਹੱਤਵਅਤੇਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ।

(ੳ) ਇਸਭਾਗਵਿਚਗੁਰਮੁੱਖੀਲਿਪੀਦੇਮਹੱਤਵਤੇਸਾਰਥਕਤਾਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਅ) ਇਸਭਾਗਵਿਚਭਾਸ਼ਾਅਤੇਉਪ-ਭਾਸ਼ਾਦੇਅੰਤਰ-ਸੰਬੰਧਤੇਨਿਖੇੜੇਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ੲ) ਇਸਭਾਗਵਿਚਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਸਯੂਨਿਟਵਿਚਕੁੱਲ 11 ਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਪਹਿਲੇਚਾਰਸਵਾਲਾਂਵਿਚੋਂ ਦੋ ਕਰਨੇਹੋਣਗੇ ਜੋ ਪੰਜ-ਪੰਜਅੰਕਾਂਦੇਹੋਣਗੇ। ਬਾਕੀ ਦਸ ਲਘੂਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂਪੰਜਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਕਿ ਦੋ-ਦੋ ਅੰਕਾਂਦੇਹੋਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁਲਅੰਕ 20 ਹੋਣਗੇ।

ਨਿਰਧਾਰਤ ਪੁਸਤਕਾਂ

1. ਵਰ-ਘਰ, ਈਸ਼ਵਰਚੰਦਰਨੰਦਾਦਾਨਾਟਕ , ਲਾਹੌਰਬੁੱਕਸ਼ਾਪ, ਲੁਧਿਆਣਾ, 1930
2. ਪੰਜਾਬੀਨਾਟਕਦਾਇਤਿਹਾਸ, ਸਤੀਸ਼ਕੁਮਾਰਵਰਮਾਂ (ਡਾ.), ਪੰਜਾਬੀਸਾਹਿਤਅਕਾਦਮੀ, ਦਿਲੀ, 2005
3. ਪੰਜਾਬੀਨਾਟਕਅਤੇਰੰਗਮੰਚ, ਕਮਲੇਸ਼ਉੱਪਲ, ਪੰਜਾਬੀਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ, 2004
4. ਕਾਲਜਪੰਜਾਬੀਵਿਆਕਰਣ, ਡਾ.ਹਰਕੀਰਤਸਿੰਘਅਤੇਗਿਆਨੀਲਾਲਸਿੰਘ, ਪੰਜਾਬਸਟੇਟਯੂਨੀਵਰਸਿਟੀਟੈਕਸਟਬੁੱਕਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
5. ਪੰਜਾਬੀਭਾਸ਼ਾਦਾਵਿਆਕਰਨ, ਜੋਗਿੰਦਰਸਿੰਘਪੁਆਰਅਤੇਹੋਰ, ਪੰਜਾਬੀਭਾਸ਼ਾਅਕਾਦਮੀ, ਜਲੰਧਰ 1999
6. ਪੰਜਾਬੀਭਾਸ਼ਾਦਾਵਿਆਕਰਨ, ਜੋਗਿੰਦਰਸਿੰਘਪੁਆਰਅਤੇਹੋਰ, ਪੰਜਾਬੀਭਾਸ਼ਾਅਕਾਦਮੀ, ਜਲੰਧਰ 1997
7. ਪੰਜਾਬੀਵਿਆਕਰਨ: ਸਿਧਾਂਤਅਤੇਵਿਹਾਰ, ਬੂਟਾਸਿੰਘਬਰਾੜ (ਡਾ.), ਚੇਤਨਾਪ੍ਰਕਾਸ਼ਨ, ਲੁਧਿਆਣਾ

MAT-SEC-156: Working with Mathematical Softwares
(Mathematica/Matlab/Octave/Maple)

Credits: 3 (L=0; T=0; P=3) Total hours: 90

(Practicals:90)Total Marks: 75 (Including Internal Assessment= 15)

Time Allowed for Examination: 3hrs.

Instructions for the Candidates and Paper Setters

Candidates will be asked to attempt five questions out of nine, carrying equal marks. Question No.1

Spread over the whole syllabus will be compulsory.

There will be two questions from each unit and the students will have to attempt one from each unit.

Course Outcome: Basic knowledge of software Mathematica, Matlab, Octave and/or Maple.

List of Practicals

Unit-I

1. Matrix operations (addition, multiplication, inverse, transpose, determinant and rank).
2. Plotting of graphs of polynomials of degree 4 and 5, the derivative graph, the the second derivative graph and comparing them.
3. Plotting functions e^{ax+b}; log(ax + b); 1=(ax + b); sin(ax + b); cos(ax + b); jax + bj and to illustrate the eff ect of a and b on the graph.
4. Obtaining surfaces of revolution of curves.

Unit-II

1. Tracing of conics in Cartesian coordinates/ polar coordinates. 3D plots.
2. Computation of limits, derivatives and integration of vector functions.
3. Tangent planes to surfaces at a given point.
4. Sketching ellipsoid, hyperboloid of one and two sheets, elliptic cone, elliptic, paraboloid, hyperbolic paraboloid using Cartesian coordinates.

Unit-III

1. Plotting the solutions of first order differential equation.
2. Plotting the solutions of second order differential equation.
3. Plotting the solutions of third order differential equation.

Unit-IV

1. Solution of initial value problems.
2. Solution of boundary value problem.
3. Exponential growth model.
4. Exponential decay model.
5. Limited growth of population.

Essential Textbooks

(A) George B. Thomas, Maurice D. Weir and Joel R. Hass, Thomas' Calculus, 12th Ed., Pearson Education, 2014.

(B) Belinda Barnes and Glenn R. Fulford, Mathematical Modeling with Case Studies, A Di erentialEquation Approach using Maple and Matlab, Second Edition, Taylor and Francis group, London and 2009.

r(dC) Martha L. Abell and James P. Braselton, Di erential Equations with MATHEMATICA, 3 ed., Elsevier Academic Press, 2004.

Further Readings

1. Joseph L. Taylor Foundations of Analysis, Pure and Applied Undergraduate Texts, 18, American Mathematical Society, Providence, RI, 2012.
2. Shanti Narayan, Integral Calculus, S. Chand and Company Ltd, 2001.
3. M.J. Strauss, G.L. Bradley and K. J. Smith, Calculus, 3rd Ed., Dorling Kindersley (India) P.Ltd. (Pearson Education), 2007.
4. H. Anton, I. Bivens and S. Davis, Calculus, 7th Ed., John Wiley and Sons, 2002.
5. R. Courant and F. John, Introduction to Calculus and Analysis (Volumes I & II), Springer-Verlag, Inc., 1989.
6. S. L. Ross, Di erential Equations, 3rd ed., John Wiley and Sons, India 2004.

7. E. A. Coddington, An introduction to ordinary di erential equations, Prentice- Hall of India, 1961.

8. C .H. Edwards and D. E. Penny, Di erential Equations and Boundary Value problemsComputing and Modeling,Pearson Education India, 2005.

9. W. E. Boyce and R. C. Diprima, Elementary di erential equations and boundary valueproblems, Seventh Edition, John Wiley and Sons, Inc., 2001

10. Earl D. Rainville and P. E. Benediet, Elementary di erential equations, Seventh Edition,Macmillian, Publishing Company, 1989.

Syllabi of B.A. (Hons.) Semester- IV (under NEP)

ECO DSC 8: INTERMEDIATE MICROECONOMICS – II

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment20 marks**
Time	**3 hours**

Course Description

This course is a sequel to Intermediate Microeconomics I. The emphasis will be on giving conceptual clarity to the student coupled with the use of mathematical tools and reasoning. It covers general equilibrium and welfare, imperfect markets and topics under information economics.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks andcontinuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT- I

Imperfect Competition

Monopoly; pricing; measurement of market power; Multiplant monopoly

Price discrimination; Inter temporal; peak-load pricing; two-part tariff

Regulation of Monopoly

Monopolistic competition: price and output determination Economic efficiency and resource allocation

UNIT-II

Imperfect Competition

Oligopoly; Price leadership; Cournot , Stackleberg and Bertrand Models

Game theory and competitive strategies: Dominant Strategy; Mixed strategies; Prisoners Dilemma

; Repeated and Sequential games

UNIT- III

General Equilibrium, Efficiency and Welfare Equilibrium and efficiency under pure exchange and production; overall efficiency Welfare economics; Social welfare, welfare maximisation, envy and equity,

UNIT- IV

Market Failure

Externalities; Production and Consumption, Coase Theorem, Market Signals, Public Goods: Provision of Public Goods, Free Riding, Different Levels of Public Goods.Asymmetric Information: Market for Lemons, Moral Hazard, Adverse Selection, Signaling.

Readings:

1. Scope as in Varian. H.R (2010). *Intermediate Microeconomics, a Modern Approach* (8thed.)W.W. Norton and Company/Affiliated East-West Press (India) given as follows:

   Unit 1 chapters 24, 25
   Unit 2 chapters 27, 28
   Unit 3 chapters 31, 32, 33
   Unit 4 chapters 34, 36, 37

Advanced Readings:

1. Varian. H.R (2010). *Intermediate Microeconomics, a Modern Approach* (8th ed.)W.W. Norton and Company/Affiliated East-West Press (India).The workbook by Varian and Bergstrom could be used for problems.
2. C. Snyder and W. Nicholson.(2010).*Fundamentals of Microeconomics,* Cengage Learning (India).
3. Maddala, G.S. and Miller, R. L. (2006). *Intermediate microeconomics: theory, issues, applications*. McGraw-Hill Companies.
4. Koutsoyiannis, A. (2003). *Modern microeconomics.(2nd ed.).* Paperback
5. B. Douglas Bernheim and Michael D. Whinston. (2009). Microeconomics Tata McGraw-Hill(India).
6. Bergstrom, T. C., & Varian, H. R. (1990). Workouts in intermediate microeconomics. WWNorton.
7. Pindyck, Robert S., and Daniel L. Rubinfeld. Microeconomics. Upper Saddle River, N.J.:Pearson/Prentice Hall, 2009.
8. Mankiw, G. (2000). Microeconomics. In *Principles of Microeconomics* (Vol. 1, pp. 3–6, 270–272).

ECO DSC 9- INTERMEDIATE MACROECONOMICS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregated analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "Intermediate Macroeconomics" equips the students at the undergraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Course Outcomes

After completion of the course, a student must be capable of understanding the determinants of money demand and supply. Also the paper shall inculcate the abilities among students to develop open economy IS-LM based models to work out the key macroeconomic targets. An understanding shall be developed on behaviour of general price level and its use as a key policy instrument to target real variables such as unemployment and output. The theories of explaining trade cycles shall make a student capable of understanding nature and causes of fluctuations in economic activity.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each).

   It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).
4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry 15 marks ($15 \times 4 = 60$).

Unit-I

(Money Supply and Demand)

***Supply of Money*:** Theoretical Debate and Empirical Attempts to define money; Components of Supply of Money, Credit Creation by Commercial Banks, Money Multiplier.

***Demand for Money*:** Classical Quantity Theory, Keynesian Theory, Baumol and Tobin's Contributions. Friedman's Restatement of Quantity Theory of Money.

Unit-II

(Simultaneous Equilibrium & Tatonnement Process under Closed and Open Economies)

IS and LM Framework: *Derivation, Properties, Shifts and Rotations of IS and LM Curves under closed and open economy systems. Derivation, Properties, Shifts, and Rotations of BP Curve. Simultaneous Equilibrium in Money and Product Markets. Impact of Opening-up on simultaneous equilibrium (i.e., IS-LM-BP simultaneous equilibrium).*

Monetary and Fiscal Policies: *Objectives, Conflicts among Objectives. Relative Effectiveness of Monetary and Fiscal Policies under Different Situations in IS-LM-BP Framework. International Trinity and Quadrilemma choices under IS-LM-BP framework.*

Unit-III

(Inflation Theories and Philips Curve)

Inflation: *Effects of Inflation. Theories of Inflation: Quantity Theory, Keynesian Theory, Monetarist views on Inflation, Modern theory of Inflation, Structural Theory.*

The menu of policy choices: *Philips Curve Analysis –Short Run and Long Run views. The Monetarist-Keynesian Debate and the Phillips Curve.*

UNIT-IV

(Trade Cycle Models/Theories)

Trade Cycle: *Hansen–Samuelson Accelerator–Multiplier Interaction Model, Hicks Model, Kaldor Model, Goodwin model of endogenous cycles.*

Recommended Readings:

Ackley, G. (2007). Macroeconomic Theory***. Macmillan.***

Allen, R.G.D. (1967). Macro-Economic Theory: A Mathematical Treatment***. Palgrave Macmillan.***

Beckerman, W. (1980). Introduction to National Income Analysis, *Littlehampton Book Services Ltd.*

Blanchard, O. (2009). Macroeconomics***. New Delhi: Pearson Education Inc.***

Blanchard, O.J. & Fischer. (1996). Lectures on Macroeconomics***, Delhi: PHI Learning Pvt.***

Ltd. Branson, W. H. (1989). Macro Economic Theory and Policy ***(3rd ed.). Harper Collins.***

Cooper, R., Edey, Harold C., and Peacock, A.T. (2003). National Income and Social Accounting.

Routledge.

D'Souza, E. (2008). Macroeconomics. ***Pearson Education.***

Dornbusch, R., Fisher, S & Startz, R. (2002). Macroeconomics***, (9th ed.). New Delhi: Tata***

McGraw Hills.

Levacic, R. & Rebmann, A. (1982). Macroeconomics: An Introduction to Keynesian Neoclassical Controversies***. Macmillan.***

Miller, R. L.& Pulsinelli, R. (1986). Macroeconomics. ***New York: Harper & Row.***

Ola Olsson (2010). Essentials of Advanced Macroeconomic Theory***, Routledge.***

Romer, D. (1996). Advanced Macroeconomics***. McGraw-Hill.***

Uma Datta Roy Choudhury (2000).National Income Accounting, *Macmillan*.

Further Readings:

Abel, A. B., Bernanke, B. S. (2007). Macroeconomics. ***New Delhi: Pearson Education Inc.***

Dornbusch, R. (1980). Open Economy Macroeconomics***. New York: Weltwirtschaftliches Archive.***

Frisch, H. (1983). Theories of Inflation***. Cambridge: Cambridge University Press.***

Froyen, R. T. (2002). Macro Economics. ***New Delhi: Pearson Education Inc.***

Gordon, R. J. (2012). Macroeconomics***. Delhi: PHI Learning Pvt. Ltd.***

Heijdra, B. J. (2017). The Foundations of Modern Macroeconomics***. Oxford University Press.***

Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle ***Oxford*****:** Clarendon Press***.***

Lucas, R. (1981). Studies in Business Cycle Theory***, Massachusetts: MIT Press Cambridge.***

Makin, A. J. (2002). International Macroeconomics***. Princeton Hall.***

Mankiw, N. G. & Romer, D. (1991). New Keynesian Economics. MIT, 2 Volumes.

Mathews, R. C. O. (1964). The Trade Cycle. ***Cambridge: Cambridge University Press.***

Pentecost, E. (2000). Macroeconomics – An Open Economy Approach. ***MacMillan Press Ltd.***

Raghbendra, J. (1964). Contemporary Macroeconomics: Theory and Policy. ***New Age International Publishers.***

Sherffrin, S. (1999). Rational Expectations. ***Cambridge University Press.***

Snowden, B & Vane, H. R. (2005). Modern Macroeconomics: Its Origin, Development and Current Stat., ***UK: Edward Press.***

Wendy, Carlin/Soskice, D. (2007). Macroeconomics: Imperfections, Institutions and Policies***. Oxford University Press.***

ECO DSC 10 – Introductory Econometrics

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Description

Objective: Application of economic theory need a reasonable understanding of economic relationship and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equations models.

Besides the time-tested method of imparting verbal instructions through lectures, description and derivation of econometric models would be demonstrated by extensive use of blackboard. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students writing assignments and written examinations.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours. The paper-setter must put a note in the question paper in this regard.
- There shall be **9** questions in all. The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20) Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

Unit-I

Introduction: Nature and Scope of Econometrics, Methodology of Econometrics.

Specification of an econometric model. Nature of Data used in estimation.

Simple Regression Models: Estimation of model by method of Ordinary Least Squares; Properties of the Least Squares estimators, the Gauss-Markov theorem. Goodness of fit -R Square & Adjusted R Square. Testing the significance of Regression Coefficients.

UNIT -II

Multiple Regression Models: Estimation of model by method of Ordinary Least Squares; Properties of the Least Squares estimators, the Gauss-Markov theorem.

Specification Errors: Types of Specification errors. Consequences and Remedial measures for Omission of a relevant variable; and Inclusion of irrelevant variable.

UNIT- III

Errors of Measurement: Errors of Measurement in Dependent and Independent variables.

Encompassing and Non-Encompassing Models: Introduction. Model Selection Criteria.

Unit-IV

Econometrics Problems: Nature, Consequences, Detection and Remedial measures of the problems of Multicollinearity, Heteroscedasticity and Autocorrelation.

Essential Reading:

Gujarati Damodar N. (2004): Basic Econometrics (4th ed.), Singapore. McGraw Hills Co.

Further Readings:

Green, William (2000): Econometric Analysis (4th Ed.), Pearson Education.

Johnston, J & Dinardo, J. (2007): Econometric (4th Ed.), McGraw Hills Co.

Maddala, G S (2007): Introduction to Econometrics, 3rd Edition, John Wiley & Sons, Ltd., England.

Pindyck, R S & Rubinfeld, D L (1998): Econometric Models & Economic, Forecasts, McGraw Hills Co.

Intriligator, M J & Bodkin, H (1996): Econometric Models Techniques and Application. Prentice Hall.

ECO DSC 11- GROWTH MODELS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective

A continuous economic growth is the primary condition for economic development. The determinants of growth differ on the basis of the stage of economic development a country faces. Thus, a study of the macroeconomic Growth Models is necessary to understand the determinants of economic growth. The idea and analysis of growth has therefore, assumed such a great significance in recent times that a prior understanding of growth theories is considered essential for the proper comprehension of the different macroeconomic issues and policies. Growth models now are not only a scientific method of economic analysis; but also a body of empirical economic knowledge.

The paper entitled "Growth Models" equips the students at the undergraduate level to understand systemic facts and latest theoretical developments for theoretical and empirical analysis of growth dynamics.

The students will be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts will be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Course Outcomes

The knowledge of theories and growth models would help the students better understand the entire economic system and its growth over time. The basic models mainly focus on the contribution of inputs to production while assuming the technology is constant. The course also covers measuring the level of technology and its role in changing economic structure. The importance of main fundamentals such as public infrastructure, institutions, and efficient resource allocation has also been covered to understand current development issues. Overall, this course would lay the foundation for the students to understand the real issues in economic development.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be short answer type word limit (25-30 each).

It would carry 15 short questions, spread over the entire syllabus. The candidate will be required

to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10 × 2 = 20).
4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry 15 marks (15 × 4 = 60).

Unit-I

(Initial Discussion on Growth Theory)

Constant capital output ratio based growth models: Domar model of growth, Harrod Model, Similarities between two models, Stability property of Harrod and types of growth, Deviations from knife edge and possible mismatches among three types of growth rates.

Unit-II

(Production Function and Growth Theories)

Neo-Classical Model including Solow Model; conditions of steady state, deviation from steady state. Solow residual as technological progress, the Golden rule. **Overlapping Generations and Dynamic Efficiency**

Unit-III

(Economic Growth and Convergence)

The Convergence Hypothesis; Beta and *sigma* convergences, idea and empirical evidences. Some advance measures of convergence; Club and Gamma convergences, some empirical evidences.

Unit-IV

(Endogenous Growth)

Limitations of neoclassical growth theories and need of endogenising growth. The AK model as initial endogenous growth model. Neoclassical endogeneous growth; capital accumulation, externalities and human capital. **Directed Technical Change: Endogenous Skill-Bias and Endogenous Labor-Augmenting Technological Change. Competition and technological change.**

Recommended Readings:

Acemoglu, Daron (2009). *Introduction to Modern Economic Growth*. Princeton University Press, 2009. ISBN: 9780691132921.

Freitas, M.L. (2022). *Economic Growth Models: A Primer / Student's Guide*, Nova School of Business and Economics, Online Available: https://sweet.ua.pt/afreitas/growthbook/MLFGrowthNotesnet.pdf

Jones, Hywel G. (1976). *An Introduction to Modern Theories of Economic Growth*, McGraw-Hill, Republished in 1976.

Solow, Robert (1970). Growth Theory: An Exposition, Clarendon Press, Oxford, UK.

From session 2024-25

Programme Outcome: The aim of the course is to pay a special attention to applications of Probability Theory. The main focus of the course is to make the students familiar with various probability distributions and their uses in real life. The focus will be both on theoretical as well as practical approach.				
Course Outcome: On completion of this course students will be able to understand the concept of probability theory, random variable and their significance in real world. Students will get knowledge about various discrete and continuous distributions to solve problems. Students will understand the usage of Central Limit Theorem.				
Class	**Sem.**	**Course Code**	**Course Title**	**Credits= 6**
B.Sc. (Honours)	**II**	**STAT-M-162**	**INTRODUCTORY PROBABILITY**	**L=4** **P=2**
Total Marks =150 (Theory= 100 + Practical= 50) Internal Assessment= 20%				
Total Hours= 120 hours (Theory= 60 hours +Practical= 60 hours)				
Time Allowed for Examination: Theory= 3 hours; Practical= 3 hours				
Instructions for Paper setter	There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit			
Instructions for Candidate	The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each Unit**.**			

Course Contents:

UNIT I

Probability: Introduction, random experiments, sample space, events, independence, dependent, mutually exclusive, exhaustive, equally likely and algebra of events. Classical, statistical, and axiomatic approaches to probability, Conditional Probability, laws of addition and multiplication, theorem of total probability, Bayes' theorem and its applications.

Random Variables (for univariate distribution only): Discrete and continuous random variables, Probability mass function (p.m.f.), Probability density function (p.d.f.), Cumulative distribution function (c.d.f.) Illustrations of random variables and their properties. Expectation, variance, moments and moment generating function.

UNIT II

Standard probability distributions: Binomial, Poisson, geometric, negative binomial, hyper geometric, uniform, normal, exponential, beta, gamma and their applications. Fitting of Binomial, Poisson and Normal Distributions Chebyshev's inequality ,Convergence in probability, Weak law of large numbers, Convergence in Distribution, De- Moivre Laplace and Lindeberg-Levy Central Limit Theorems (C.L.T.).

1. Hogg, R.V., Tanis, E.A. and Rao J.M. (2009): Probability and Statistical Inference, Seventh Ed, Pearson Education, New Delhi.

2. Miller, Irwin and Miller, Marylees (2006): John E. Freund's Mathematical Statistics with Applications, (7th Edn.), Pearson Education, Asia.

3. Rohatgi, V. K. and Saleh A.K.M.E. (2010) : An Introduction to Probability Theory and Mathematical Statistics, Wiley Eastern.

4. Meyer, P.L. (1970): Introductory Probability and Statistical Applications, Oxford & IBH Publishing, New Delhi .

STAT-M-162: INTRODUCTORY PROBABILITY (PRACTICAL)

Practical: Max. Marks: 50 Practical will be based on Paper STAT-M-162: INTRODUCTORY PROBABILITY.

Practical: Final Written Examination: 30 marks, Internal Assessment: 10 marks, viva-voce: 10 marks

Note for Practical: The practical question paper will consist of 5 questions. The students will be required to attempt any three questions in three *hours* duration.

Compulsory Punjabi
ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ
PUN-128
FYUGPG B.A (Hons.) in Economics/B.A (Hons) in Economics (Research)

Semester- IV **ਕ੍ਰੈਡਿਟ-2**

Credit Distribution, Eligibility and Pre-Requisites of the Course

Course title & Code	Credits	Credit distribution of the Course			Eligibility criteria	Pre-requisite of the course (if any)
		Lecture	tutorial	practical		
PUN-128	2	3	1	0		

Credits: 2 **L/T/P=3/1/0**

Learning objective

- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਕਹਾਣੀਸੰਬੰਧੀਮੁੱਢਲਾਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਕਹਾਣੀਦੇਵਿਸ਼ੇਤੇਕਲਾਪੱਖਸੰਬੰਧੀਮੁੱਢਲਾਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਨਿਬੰਧਲੇਖਣਦੇਹੁਨਰਨੂੰਗ੍ਰਹਿਣਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਵਿਹਾਰਕਸੂਚਨਾਹਿਤਨੋਟਿਸਤੇਮੁਹਾਵਰਿਆਂਦੀਵਰਤੋਂਸੰਬੰਧੀਜਾਣਕਾਰੀਪ੍ਰਾਪਤਕਰਨਗੇ।

Learning outcomes

- ਇਸਨਾਲਵਿਦਿਆਰਥੀਅੰਦਰਪੰਜਾਬੀਕਹਾਣੀਲੇਖਣਸੰਬੰਧੀਮੁੱਢਲੀਸੂਝਪੈਦਾਹੋਵੇਗੀ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਪੰਜਾਬੀਕਹਾਣੀਦੇਵਿਸ਼ਿਆਂਨੂੰਸਮਝਣਦੇਯੋਗਬਣਨਗੇ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਨਿਬੰਧਲੇਖਣਦੀਬਣਤਰਨੂੰਸਮਝਣਦੇਯੋਗਬਣਨਗੇ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਵਿਹਾਰਕਵਰਤੋਂਵਿਚਆਉਣਵਾਲੇਸੂਚਨਾਹਿੱਤਨੋਟਿਸਤੇਮੁਹਾਵਰਿਆਂਦੀਵਰਤੋਂਸੰਬੰਧੀਸੂਝਪ੍ਰਾਪਤਕਰਪਾਉਣਗੇ।

ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ
ਬੀ.ਏ ਅਰਥ ਸ਼ਾਸਤਰ (ਆਨਰਜ਼)
PUN-128
Compulsory Punjabi
FYUGPG B.A (Hons.) in Economics/B.A (Hons) in Economics (Research)

Semster:IV

ਕ੍ਰੈਡਿਟ–2
ਨਿਰਧਾਰਤਕੋਰਸ ਸਮਾਂ:60 ਘੰਟੇ
ਕੁਲ ਅੰਕ:50
ਲਿਖਤੀ ਅੰਕ:40
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ:10

(ਇਸਕੋਰਸਵਿਚ 2 Unit ਹੋਣਗੇ। Unit-1 ਵਿਚਨਿਰਧਾਰਿਤਕਹਾਣੀਸੰਗ੍ਰਹਿ 'ਕਥਾ–ਕਹਾਣੀ' ਵਿੱਚੋਂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ। ਯੂਨਿਟ ਦੋ ਵਿਚਵਿਹਾਰਕਪੰਜਾਬੀਨਾਲਸੰਬੰਧਿਤਸਵਾਲਪੁੱਛੇਜਾਣਗੇ, ਜੋ 20 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।)

Unit-1

ਪੁਸਤਕ 'ਕਥਾ–ਕਹਾਣੀ' (ਕਹਾਣੀਸੰਗ੍ਰਹਿ), ਡਾ. ਧਨਵੰਤਕੌਰ (ਸੰਪਾ.), ਪਬਲੀਕੇਸ਼ਨਬਿਊਰੋ, ਪੰਜਾਬਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

(ੳ) ਕਹਾਣੀਦਾਵਿਸ਼ਾ–ਵਸਤੂਤੇਮੰਤਵ (ਦੋ ਵਿਚੋਂਇੱਕ)

(ਅ) ਕਹਾਣੀਦੇਕਿਸੇਪਾਤਰਦਾਪਾਤਰ–ਚਿਤ੍ਰਣ(ਦੋ ਵਿਚੋਂਇੱਕ)

(ੲ) ਪੁਸਤਕਵਿਚੋਂਲਘੂਪ੍ਰਸ਼ਨ: ਕਹਾਣੀਆਂਤੇਲੇਖਕਾਂਦੇਸੰਬੰਧਵਿਚ (ਦਸ ਵਿਚੋਂਪੰਜ)

(5+5+10=20 ਅੰਕ)

Unit-2ਵਿਹਾਰਕਪੰਜਾਬੀ

(ੳ) ਨਿਬੰਧਲੇਖਣ: ਪੰਜਾਬੀਸਭਿਆਚਾਰਨਾਲਸੰਬੰਧਿਤ (ਦੋ ਵਿਚੋਂਇੱਕ)

(ਅ) ਮੁਹਾਵਰਿਆਂਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂ(ਸੱਤਵਿਚੋਂਪੰਜ)

(ੲ) ਅਖਾਣਾਂਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂ (ਸੱਤਵਿਚੋਂਪੰਜ)

(10+5+5 =20 ਅੰਕ)

ਨੋਟ: ਲਾਜ਼ਮੀਪੰਜਾਬੀਵਿਸ਼ੇਦੇਕ੍ਰੈਡਿਟ 2 ਹੋਣਗੇਅਤੇਪੇਪਰ 50 ਅੰਕਾਂਦਾਹੋਵੇਗਾ। ਵਿਦਿਆਰਥੀਹਫ਼ਤੇਵਿਚਘੱਟੋ–ਘੱਟਪੰਜਦਿਨਪੜ੍ਹਨਗੇ।

ਪੇਪਰਸੈਟਰ/ਅਧਿਆਪਕਲਈਹਦਾਇਤਾਂਅਤੇਪ੍ਰਸ਼ਨ–ਪੱਤਰਦੀਰੂਪ–ਰੇਖਾ:–

1. ਪ੍ਰਸ਼ਨ–ਪੱਤਰਦੇਕੁੱਲ 50 ਅੰਕਹਨ ਜੋ ਕਿ 2 ਕ੍ਰੈਡਿਟਦੇਬਰਾਬਰਹੈਅਤੇਕੋਰਸਦਾਨਿਰਧਾਰਤਸਮਾਂ 60 ਘੰਟੇਹੋਵੇਗਾ।
2. ਲਿਖਤੀਪ੍ਰਸ਼ਨ–ਪੱਤਰ 40 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।
3. ਇੰਟਰਨਲਅਸੈਸਮੈਂਟਦੇ 10 ਅੰਕਹੋਣਗੇ।

4. ਪੇਪਰਦਾਸਮਾਂਤਿੰਨਘੰਟੇਹੋਵੇਗਾ।

ਯੂਨਿਟਅਤੇਅੰਕਾਂਦੀਵੰਡ

1. ਯੂਨਿਟਇਿਕਵਿਚ 'ਕਥਾ-ਕਹਾਣੀ' (ਕਹਾਣੀਸੰਗ੍ਰਹਿ) ਪੁਸਤਕਵਿਚੋਂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਇਹਯੂਨਿਟ 20 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।

(ੳ) ਇਿਸਭਾਗਵਿਚਕਹਾਣੀਦੇਵਿਸ਼ਾ-ਵਸਤੂਅਤੇਮੰਤਵਬਾਰੇਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਅ) ਇਿਸਭਾਗਵਿਚਪਾਤਰ-ਚਿਤਰਣਬਾਰੇਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ੲ) ਇਿਸਭਾਗਵਿਚਕਹਾਣੀਆਂਅਤੇਕਹਾਣੀਲੇਖਕਾਂਸੰਬੰਧੀਲਘੂਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਿਸਯੂਨਿਟਵਿਚੋਂਕੁਲ 16ਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਪਹਿਲੇਚਾਰਿਸਵਾਲਾਂਵਿਚੋਂ ਦੋ ਕਰਨੇਹੋਣਗੇ,ਜਿਸਵਿਚੋਂਹਰ ਿਸਵਾਲਪੰਜਅੰਕਾਂਦਾਹੋਵੇਗਾ।ਬਾਕੀਸੱਤਲਘੂਿਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂਪੰਜਿਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਿਕ ਦੋ-ਦੋ ਅੰਕਦੇਹੋਣਗੇ।ਇਿਸਯੂਨਿਟਦੇਕੁਲਅੰਕ 20 ਹੋਣਗੇ।

2. ਯੂਨਿਟ ਦੋ ਵਿਚਪੰਜਾਬੀਸਭਿਆਚਾਰਨਾਲਸੰਬੰਧਤਨਿਬੰਧਲੇਖਣ, ਮੁਹਾਵਰਿਆਂਅਤੇਅਖਾਣਾਂਬਾਰੇਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਇਿਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ।

(ੳ) ਇਿਸਭਾਗਵਿਚਪੰਜਾਬੀਸਭਿਆਚਾਰਨਾਲਸੰਬੰਧਤਨਿਬੰਧਲਿਖਣਬਾਰੇਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਅ) ਇਿਸਭਾਗਵਿਚਮੁਹਾਵਰਿਆਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂਸੰਬੰਧੀਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ੲ) ਇਿਸਭਾਗਵਿਚਅਖਾਣਾਂਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂਬਾਰੇਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਿਸਯੂਨਿਟਵਿਚਕੁੱਲ 16ਿਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਪਹਿਲੇਦੋਿਸਵਾਲਾਂਵਿਚੋਂਇਿਕਸਵਾਲਹੱਲਕਰਨਾਹੋਵੇਗਾ ਜੋ ਦਸਅੰਕਾਂਦਾਹੋਵੇਗਾ।ਬਾਕੀ14 ਲਘੂਿਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂ10ਿਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਿਕਇਿਕ-ਇਿਕਅੰਕਾਂਦੇਹੋਣਗੇ।ਇਿਸਯੂਨਿਟਦੇਕੁਲਅੰਕ 20 ਹੋਣਗੇ।

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